MDC CLUB MANAGEMENT SYSTEM

PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF

BACHELOR OF COMPUTER APPLICATIONS

To

MARIAN COLLEGE KUTTIKKANAM [AUTONOMOUS]

Affiliated to

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

By

ADITHYAN H

(Reg.No:23UBC203)

MELBIN ANTONY

(Reg.No:23UBC241)

Guided By

Ms. DIVYA LAKSHMI S

DEPARTMENT OF COMPUTER APPLICATIONS

MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

PEERMADE – 685531

OCTOBER, 2025

DECLARATION

We, **ADITHYAN H [Reg.no 23UBC203], MELBIN ANTONY [Reg.no 23UBC241]** certify that the Mini project report entitled **"MDC CLUB MANAGEMENT SYSTEM"**is an authentic work carried out by us at Marian College Kuttikkanam(Autonomous). The matter embodied in this project work has not been submitted elsewhere for the award of any degree or diploma to the best of our knowledge and belief.

Signature of the Students:

ADITHYAN H

MELBIN ANTONY

Date:

BONAFIDE CERTIFICATE

This is to certify that this project work entitled **"CELESTIA INTERIOR & EXTERIOR DESIGNS"** is a bonafide record of work done by **Ms. SHEETHAL S [Reg.no 22UBC257], Mr. DERIN JACOB BABU [Reg.no 22UBC222**] at Marian College Kuttikkanam (Autonomous) in partial fulfilment for the award of the **Degree of Bachelor of Computer Applications** of **Mahatma Gandhi University, Kottayam.**

This work has not been submitted elsewhere for the award of any degree to the best of our knowledge.

Head of the Department	Internal Guide
Dr Rajimol A	**Ms. Divya Lakshmi S**
Dept. of Computer Applications	Dept. of Computer Applications
Marian College Kuttikkanam	Marian College Kuttikkanam
Peermade – 685531	Peermade - 685531

Submitted for the Viva-Voice Examination held on

DEPARTMENT SEAL

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

"Gratitude is a feeling which is more eloquent than words, more silent than silence." In undertaking this project, we needed the direction, assistance and cooperation of various individuals and organizations, which is received in abundance with grace of God, without their unconstrained support, the project could not have been completed. If words are considered as the symbol of approval and token of acknowledgement, then let the following words play the heralding role of expressing our gratitude.

We wish to acknowledge our sincere gratitude to our Manager, **Rev.Fr. Boby Alex Mannamplackal** and **Dr Ajimon George**, Principal **Marian College Kuttikkanam (Autonomous),** for all their efforts and administration in educating us in their premier institution.

We extend our gratitude to **Dr Rajimol A**, Head of the Department of Computer Application, who is a constant source of inspiration and whose advice helped us to complete this project successfully.We express our deep sense of gratitude to our internal project guide, **Ms Divya Lakshmi S**, for her profound guidance for the successful completion of this project. With great enthusiasm we express our gratitude to all the faculty members of Department of Computer Applications for their timely help and support.

Finally, we express our deep appreciation to all our friends and family members for the moral support and encouragement they have given to complete this project successfully.

Melbin Antony, Adithyan H

ABSTRACT

ABSTRACT

The project named **“MDC Club Management System”** is a web-based platform designed to manage, organize, and monitor the activities of the MDC Club in an efficient digital environment. The system streamlines club operations by providing essential tools for students, faculty, and administrators, ensuring smooth communication, event participation, merchandise management, and academic support.

The platform consists mainly of three roles: **Students, Faculty, and Admins**.

- **Students** can register, log in, participate in events, purchase club merchandise through a dummy payment gateway, view their purchase history, confirm deliveries, check attendance status, and download participation certificates if eligible. They also have access to a gallery of past events and can review their participation history.
- **Faculty** can review and approve event registrations, evaluate reports submitted by admins, approve or reject student purchases, and participate in real-time communication with admins via the chat system.
- **Admins** can create and manage events, upload media (photos and videos of programs), manage merchandise stock, track orders, take attendance for events, oversee student participation, and communicate with faculty. They are also responsible for handling event proposals and approvals.

The platform integrates several key functionalities including attendance monitoring, merchandise management, report submission and approval workflows, event proposal handling, and a secure dummy payment simulation for merchandise purchases. It ensures real-time updates and maintains detailed records of student activities, participation, and transactions for transparency and accountability.

The system uses **HTML, CSS, and JavaScript** to provide an engaging and responsive front-end interface, while the **back end is developed with PHP and MySQL**, enabling reliable database management and smooth system operations.

TABLE OF CONTENTS

INTRODUCTION

1.INTRODUCTION

1.1 ABOUT THE PROJECT

The **MDC Club Management System** is a web-based platform designed to digitalize and automate the day-to-day activities of college dance club. The system provides an integrated solution for managing events, tracking student participation, handling merchandise sales, generating attendance records, and facilitating smooth communication between students, faculty, and administrators.

Traditionally, student clubs depend on manual methods such as physical registers, paper-based proposals, or informal communication channels for conducting their activities. These approaches are often time-consuming, error-prone, and lack transparency. To overcome these limitations, the MDC Club portal centralizes all functionalities into a single, user-friendly platform.

The project mainly provides three distinct roles: **Admin, Faculty, and Student**.

- **Admins** can create events, manage registrations, approve merchandise purchases, and oversee reports.
- **Faculty** can approve or reject event proposals, review student reports, and monitor participation.
- **Students** can register for events, check participation status, purchase club merchandise with a dummy payment gateway, submit reports, and access certificates for their participation.

In addition, the system introduces features like an **attendance management module**, **report submissions**, **faculty-admin chat system**, and a **media gallery** to showcase past events. This ensures a complete end-to-end digital transformation of club activities.

1.1.1 THE PURPOSE AND SCOPE

The primary purpose of this project is to streamline club operations by replacing traditional manual processes with a robust online system. The **scope** of the system covers:

- Event creation, registration, approval, and participation monitoring.
- Attendance tracking and automated certificate generation for eligible students.
- Merchandise sales with inventory management and dummy payment gateway support (Card/UPI).

- Report submission and review workflows between Admins and Faculty.
- Communication and collaboration via integrated chat functionality.
- Media gallery for sharing past event photos and videos.

By covering these modules, the MDC Club portal ensures transparency, efficiency, and accessibility for all stakeholders. Students, faculty, and administrators can access and manage data in real time, improving the overall functioning of the club

1.2 EXISTING SYSTEM

Currently, most of the club activities are carried out manually, which makes the process inefficient and time-consuming. Event registrations are often managed using paper-based forms or informal communication channels, making it difficult to maintain accurate records. Attendance tracking is done manually, which increases the chances of errors and inconsistencies. Merchandise sales are unorganized, lacking proper stock management and payment tracking. Report submissions are scattered across emails or physical documents, leading to difficulties in maintaining centralized records. Furthermore, there is no dedicated communication platform between the Admin and Faculty, which causes delays in decision-making. Certificate distribution is also delayed due to manual verification processes, affecting the overall efficiency of the system.

1.3 PROPOSED SYSTEM

The proposed MDC Club Management System overcomes the limitations of the manual system by offering a centralized and automated platform for all club activities. The system enables students to register for events online, track their participation, and automatically receive attendance-based certificates. Merchandise purchases are streamlined with proper stock management and payment confirmation, while report submission becomes more efficient with digital uploads, faculty review, and feedback integration. The communication gap between Admin and Faculty is resolved through an inbuilt chat system, ensuring faster approvals and collaboration. Additionally, the inclusion of a digital gallery provides visibility for past events and enhances the user experience. By integrating all these features into a single web application, the proposed system ensures transparency, accountability, and improved management of club operations.

SYSTEM ANALYSIS

2. SYSTEM ANALYSIS

2.1 PROBLEM DEFINITION

The current management of college clubs and student activities is often manual or partially digital, which leads to inefficiencies in handling events, merchandise, attendance, and communication between students, faculty, and administrators. Records are frequently scattered, making tracking participation, managing orders, and approving reports cumbersome. These inefficiencies cause delays, errors, and lack of transparency in club operations.
To overcome these limitations, the **MDC Club Management System** is proposed as a web-based platform to digitally streamline all club-related activities, providing a centralized and efficient system for students, faculty, and admins.

2.1.1 ADVANTAGES OF PROPOSED SYSTEM

- **Avoids Redundancy:** The system uses validation and database constraints to eliminate duplicate records, improving overall efficiency.
- **Quick Access and Processing:** Digital management ensures faster access to event, attendance, and purchase data compared to manual systems.
- **Time Efficiency:** Automated workflows reduce time required for approvals, report submissions, and merchandise management.
- **Reduces Paperwork:** Records, certificates, and reports are managed digitally, minimizing paper usage.
- **Provides Accurate Information:** Real-time updates and validations ensure reliable and accurate data for decision-making.

2.2 FEASIBILITY ANALYSIS

Feasibility study evaluates the system's workability, its ability to meet user requirements, and the efficient use of resources. The objective is to assess whether the project is technically, operationally, and economically viable before implementation.

The feasibility study is classified into:

- Operational Feasibility
- Technical Feasibility
- Economic Feasibility

2.2.1 OPERATIONAL FEASIBILITY

The proposed system offers a user-friendly interface and faster processing speed, reducing manual effort for students, faculty, and admins. Most operations, such as event registration, report approval, attendance tracking, and merchandise orders, are automated. Therefore, the workload is minimized, and the system is operationally feasible.

2.2.2 TECHNICAL FEASIBILITY

Technical feasibility deals with hardware as well as software requirements and to what extend it can support the proposed system. The hardware required is an android phone and software is Android Studio. If the necessary requirements are made available with the system, then the proposed system is said to be technically feasible.

2.2.3 ECONOMIC FEASIBILITY

Economic feasibility is an important factor. Since the existing system is manual on the feasibility for wrong data entry is higher and consumes a lot of time and can occur errors. But the proposed system aims at processing of information’s efficiently, thus saving the time. The new system need only a system and which is already available therefore the cost is negligible. Proposed system uses validation check so there are no errors. Even though an initial investment has to be made on the software and the hardware aspects, the proposed system aims at processing of information’s efficiently. Thus, the benefits acquired out of the system are sufficient enough for the project to be undertaken. So, the proposed system is economically feasible.

2.3 RECOMMENDED IMPLEMENTATIONS

Two principle sources of data are:

- Written documents
- Data from the persons, who are involved in the operation of the system under study.

The different fact-finding techniques are:

- Questionnaires

- Personal Interviews
- Observations

Questionnaires

Questionnaires are best methods to probe data out of the customers. In this case, questionnaires were not used for data collection as the administration was small in number and they could be asked questions in a more effective interview.

Personal Interviews

Interviews were conducted with club admins and faculty to understand current processes, challenges, and expectations. Their suggestions were incorporated into the system design.

Observations

By observing current club operations, including event management, attendance recording, and merchandise handling, the workflow and data requirements for a robust database and system architecture were identified.

SOFTWARE REQUIREMENT SPECIFICATIONS

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 INTRODUCTION

Software Requirement Specification (SRS) is a critical step in the development lifecycle of the MDC Club Management System. This document defines the system functionalities required by Marian College to manage its dance club operations effectively. It is aimed at establishing a mutual understanding between stakeholders (admins, faculty, and students) and the developers, ensuring the system meets all necessary expectations with clarity and precision.

3.2 PURPOSE

The purpose of this Software Requirements Specification (SRS) is to define the scope, requirements, and structure of the **MDC Club Management System**. This web-based application provides a centralized platform to efficiently manage all club activities, including student registration, event participation, attendance tracking, announcements, report submissions, and merchandise orders. It aims to streamline communication and coordination among students, faculty, and administrators, reducing manual effort and improving transparency.

The SRS serves the following objectives:

- Establish a clear understanding between developers, faculty in-charges, and student users regarding the system's functionalities.
- Provide a basis for system validation and verification to ensure that the final product meets intended requirements.
- Reduce development cost, time, and potential errors by clearly defining expected system behavior.
- Assist stakeholders in accurately identifying and communicating their needs, ensuring that all essential features are incorporated.

In summary, this SRS acts as a roadmap for the development of the MDC Club Management System, ensuring that the system is reliable, efficient, and capable of supporting smooth management of club operations while meeting the needs of all users.

3.3 SCOPE

3.3.1 SYSTEM STATEMENT OF SCOPE

The MDC Club Management System is developed to streamline club operations at Marian College. It features a role-based login system (admin, student, faculty), event registration, announcement broadcasting, attendance marking, and a student support module. The system enhances communication and collaboration between students and coordinators while reducing manual effort.

3.4 TECHNICAL OVERVIEW

3.4.1 USER CHARACTERISTICS

There are three user types:

- **Admins**: Manage announcements, attendance, user profiles, event participation, and merchandise orders.
- **Faculty**: Track event involvement and attendance, view student participation.
- **Students**: Register and log in, view announcements, participate in events, and track orders and attendance.

3.5 FUNCTIONAL REQUIREMENTS

The functional requirements of this website are as follows:

1. **User Authentication & Registration**

- Students can register using their name, roll number, email, department, phone, and password.
- All users can log in using their credentials. Admin passwords are securely hashed for protection.

2. **Event Management**

- Admins can create, update, and manage events.
- Students can view event details and register for participation.

3. **Event Proposal Submission**

- Students or admins can submit event proposals to faculty for review and approval.
- Faculty can approve or reject proposals, ensuring proper planning and authorization.

4. **Announcement Management**

- Admins can post announcements for events, updates, or notices.
- Students and faculty can view announcements through their dashboards.

5. **Attendance Tracking**

- Admins can mark attendance during events and maintain records for each participant.
- Students can check their attendance status for events they participated in.

6. **Student & Faculty Dashboards**

- Each user role has a customized dashboard providing access to relevant tools, data, and notifications.

7. **Contact Module**

- Students can submit queries or requests through a contact form.
- Admins can view and respond to these queries via their dashboard.

8. **Media Gallery & Merchandise Management**

- Admins can upload event photos and videos to the media gallery.
- Students can view event media and order merchandise.

9. **Merchandise Delivery Tracking**

- Students can track the delivery status of purchased merchandise.
- Admins can update order and delivery status to ensure transparency.

3.6 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements of this website are as follows:

1. **Usability:**

   Clean and responsive UI, accessible via desktop and mobile.

2. **Reliability:**

   System remains stable during regular operations.

3. **Availability:**

   Accessible on LAN or server-based deployment 24/7.

4. **Security:**

   Passwords are hashed. Session-based access restricts page views.

5. **Performance:**

   Fast response to user interactions; minimal loading time.

6. **Scalability:**

   Easy to extend with new features or higher user volume.

7. **Maintainability:**

   Modular code with clear documentation.

3.7 STATED REQUIREMENTS

3.7.1 GENERAL REQUIREMENTS

The system has 7 functional modules divided between admin, faculty and students.

1. Login

- Only registered students, faculty, and admins can log in to the system to access their respective services.
- Users and admins log in using their email ID and password.
- The user ID must be a valid email address.
- Passwords can contain uppercase and lowercase letters, numbers, and special characters for security.
- Admins are redirected to the admin dashboard upon successful login.
- Students and faculty are redirected to their respective dashboards after logging in successfully.

2. Sign Up / Registration

- New students must register to access the platform's services.
- Registration requires the following fields:
  - Full Name
  - Roll Number
  - Email Address
  - Department
  - Phone Number
  - Password
  - Confirm Password

3. Admin Panel

The admin panel allows administrators to manage various processes such as:

- Creating and managing events
- Managing student participation and attendance
- Handling event proposals submitted by students or admins
- Uploading event media (photos/videos)
- Managing merchandise stock, orders, and delivery status
- Viewing queries submitted by students via the contact module

4. Event Management

- Students can browse upcoming events and register for participation.
- Admins can create, update, and manage events efficiently.
- Faculty can review and approve event proposals before confirmation.

5. Attendance Tracking

- Admins can mark and maintain attendance for events.
- Students can check their attendance status for events they participate in.

6. Media Gallery & Merchandise

- Students can view photos and videos of past events in the gallery.
- Students can order merchandise through a dummy payment gateway.
- Admins can update and track merchandise delivery status.

7. Communication & Queries

- Students can submit queries via the contact form.
- Admins can view and respond to student queries through the dashboard.
- Faculty and admins can communicate through the system for approvals and updates.

3.7.2 INPUTS

The MDC CLUB MANAGEMENT SYSTEM platform will take various inputs to provide its services:

- **User Registration:**

- Full Name
- Roll Number
- Email Address
- Department
- Phone Number
- Password and Confirm Password

 Login:

- Email Address (User ID)
- Password

 Event Participation / Proposal Submission:

- Selected Event
- Event Proposal Details (Title, Description, Date, etc.)

 Merchandise Orders:

- Selected Merchandise
- Quantity
- Delivery Address (if applicable)

 Attendance and Participation Tracking:

- Event Name
- Participant Roll Number
- Attendance Status

 Contact / Queries:

- Query Subject
- Query Description

3.7.3 PROCESSING

The system will perform the following key processing tasks:

 Data Validation:

- Validate email format and password complexity.
- Ensure all required fields are completed before submission.
- Check for duplicate entries during registration or event participation.

 Event and Proposal Management:

- Admins create, update, and manage events.
- Students submit event proposals for faculty approval.
- Faculty approve or reject proposals based on requirements.

 Attendance Management:

- Admins mark attendance for each event.
- System maintains accurate participation records.

 Merchandise Management:

- Students add items to cart and place orders via a dummy payment gateway.
- Admins update stock and track delivery status of merchandise.

 Communication Management:

- Students submit queries via contact form.
- Admins and faculty respond to queries through their dashboards.

 Reporting and Notifications:

- Generate reports on event participation, orders, and attendance.
- Notify students about event approvals, order status, and announcements.

3.7.4 OUTPUTS

The system produces the following outputs:

- **Event and Proposal Information:**

- Confirmed event registrations and approved proposals
- Event details including date, venue, and participants

- **Attendance Reports:**

- Event-wise attendance lists
- Individual student participation records

- **Merchandise and Order Details:**

- Order confirmation and payment status
- Delivery updates for purchased merchandise

- **Announcements and Notifications:**

- Notifications about new events, approvals, and updates
- Responses to student queries and messages

- **Media and Gallery Outputs:**

- Uploaded photos and videos of past events for student and faculty viewing

3.8 EXTERNAL INTERFACE REQUIREMENTS

3.8.1 USER INTERFACES

All user interfaces will be GUI interfaces, designed for ease of use and high functionality. The interfaces will have a pleasing appearance and intuitive design.

- **Design and Appearance**:
    - The interface will use suitable design elements and pleasing colors to create a comfortable and attractive environment for users.
    - Uniform design themes will be maintained across dashboards, event pages, and other modules.
    - Input controls such as textboxes, dropdowns, buttons, and checkboxes will facilitate easy data entry.
- **Usability Components**:
    - Input controls such as textboxes, dropdowns, buttons, and checkboxes will facilitate easy data entry.
    - All input fields will have clear labels and guidance to avoid errors.
    - Intuitive navigation menus and buttons will allow users to move seamlessly between dashboards, events, media gallery, and merchandise pages.
- **Responsive Design**:

    The system interface will be fully responsive, ensuring proper display and usability on desktops, laptops, tablets, and smartphones.

3.8.2 HARDWARE INTERFACES

The MDC Club Management System is a web-based platform and requires only standard computing devices with network access. No additional specialized hardware is needed.

Hardware Specification

- **Processor**: Intel Pentium or higher
- **RAM**: 256 MB or higher
- **Hard Disk Drive**: 100 MB required on disk
- **Keyboard**: Standard keyboard and mouse

Implementation Specification

- **Operating System**: Windows OS
- **Network:** Internet connection for web access

3.8.3 SOFTWARE INTERFACES

Software Specification

- **Operating System**: Windows 11
- **Web Server**: Apache or any compatible server
- **DBMS**: MySQL
- **Tool Used**: PHP, HTML, CSS, JavaScript

SYSTEM DESIGN

4. SYSTEM DESIGN

4.1 INTRODUCTION

The design phase aims to develop a solution for the problems identified during the analysis phase. For the **MDC Club Management System**, this phase transitions from understanding the requirements to creating a structured solution. System design details all necessary features and operations, including screen layouts, process diagrams, module interactions, database structures, and user interface designs.

During this phase, the overall architecture of the system is defined, including the separation of modules for Students, Faculty, and Admins. Proper design ensures that errors are minimized and system development proceeds efficiently.

Key objectives considered during design include:

- **Promptness:** The system provides clear and intuitive navigation for all users.
- **Memory Load:** Interfaces are designed with simplicity to avoid overwhelming users with too many choices.
- **Service Reachability:** Users can access all services in minimal steps to avoid frustration.
- **Navigation:** Users can easily move between dashboards, events, media gallery, and merchandise pages.
- **Error Handling:** The system provides clear error messages and validation to reduce mistakes.
- **User Updates:** Users receive notifications for event approvals, merchandise status, and announcements.

Subsystems are divided into modules, each with a clear purpose and detailed logic specifications. A modular design approach ensures maintainability, reusability, and scalability.

4.2 DESIGN METHODOLOGY

4.2.1 INPUT DESIGN

Input design focuses on converting user actions into a format recognized by the system. In the MDC Club Management System, the goal is to make data entry simple, logical, and error-free.

Key inputs include:

- **User Registration:** Name, Roll Number, Email, Department, Phone, Password
- **Login:** Email ID and Password
- **Event Registration / Proposal Submission:** Event selection, proposal details
- **Merchandise Orders:** Item selection, quantity, and delivery information
- **Attendance / Participation:** Event name, participant roll number, attendance status
- **Contact / Queries:** Subject and description

Activities in input design:

- **Data Recording:** Collecting input from users.
- **Data Verification:** Ensuring accuracy of submitted data.
- **Data Conversion:** Formatting data for processing and storage.
- **Data Validation:** Checking for errors, duplicates, and missing information.
- **Data Correction:** Promptly correcting any invalid inputs.

4.2.1 OUTPUT DESIGN

Output design ensures that the system generates meaningful and accurate information for all users. Key outputs of the system include:

- **Event and Proposal Details:** Confirmed registrations, approved proposals, event schedules
- **Attendance Reports:** Individual student participation records and event-wise attendance
- **Merchandise and Order Details:** Order confirmations, delivery status, payment receipts
- **Announcements and Notifications:** Updates for students and faculty regarding events, approvals, and queries
- **Media Outputs:** Event photos and videos accessible in the gallery

Outputs are designed to be clear, concise, and actionable, ensuring that users can easily understand system activities.

4.2.3 CODE DESIGN

The coding phase translates detailed design into executable code. The system uses a **modular approach**, with separate functions for registration, login, event management, attendance, merchandise, and reporting. Each module is named to reflect its function, making the system maintainable and easy to understand.

Key coding principles:

- Minimize code duplication through reusable functions
- Encapsulate complex operations within modules
- Implement proper error handling and data validation

4.2.4 DATABASE DESIGN

The database design of the *MDC Club Management System* is structured to efficiently handle the diverse activities of the club, such as event management, merchandise handling, attendance tracking, report submissions, and communication between users. The database was designed with the primary goal of ensuring integrity, eliminating redundancy, and enabling smooth interaction between different modules of the system.

A database system represents the underlying information system, storing integrated data in an organized manner. In the MDC Club project, the database design ensures quick, flexible, and cost-effective access to information for administrators, faculty, and students.

The design objectives of the MDC Club database include:

- Controlled Redundancy: By applying normalization techniques, duplicate data is minimized across tables, ensuring consistency and preventing anomalies.
- User-Friendly Access: Tables are structured so that queries and retrievals remain simple and intuitive for system modules.
- Data Independence: Changes to data structure at the physical level do not affect the logical representation or how users interact with the data.
- Cost-Effective Retrieval: Optimized queries allow efficient access to critical information such as participation status, order history, and event details.
- Accuracy and Integrity: Primary keys, foreign keys, and constraints are used to ensure the validity and uniqueness of data across tables.
- Failure Recovery: Proper backup and recovery strategies ensure that important data such as attendance, reports, and payments are not lost during system failures.
- Privacy and Security: User roles (admin, faculty, student) enforce controlled access to

sensitive information. For example, students can only see their records, while admins oversee complete data.

- Performance Optimization: Indexed columns and properly structured relationships improve performance during large data operations, such as retrieving participation history or generating reports.

Database design in this project incorporates both logical and physical views of data:

- The logical view represents the relationships among entities such as students, events, faculty, attendance, reports, and merchandise. It defines how data is linked conceptually (e.g., a student participates in multiple events, an order is linked to merchandise, or a report is submitted by a student and reviewed by a faculty member).
- The physical view specifies how this information is stored on the server and accessed during execution.

Each table in the MDC Club database is uniquely identified using a primary key, ensuring that records can be distinguished from one another. Foreign keys are used to establish relationships between tables, such as linking student_profiles with event_participation or connecting orders with merchandise. These relationships provide data integrity and accurate mappings across the system. Additionally, normalization techniques were applied to maintain an efficient structure. This minimizes redundancy and ensures that the database supports seamless insertion, update, and deletion operations without anomalies.

The result is a robust and secure database design that forms the backbone of the *MDC Club Management System*, enabling reliable event management, merchandise operations, attendance tracking, and communication features.

4.3 SYSTEM ARCHITECTURE AND PROCESS FLOW

UML DIAGRAMS

4.3.1 USE CASES

REGISTRATION

Use Case Id:	MDC_UC_01
Use Case Name:	Registration
Created by:	Adithyan H & Melbin Antony
Date Created:	20-09-2025
Description:	Allows new users (students, faculty, or admins) to register for an account on the MDC Club platform.
Primary actor:	User
Secondary actor:	None
Precondition:	User navigates to the registration page
Postcondition:	User account is successfully created.
Main flow:	1. User navigates to the registration page. 2. User provides required details such as roll number, name, email/phone, and password. 3. User submits the registration form. 4. System validates the information. 5. System creates a new user account and stores it in the database. 6. Use case ends.

LOGIN

Use Case Id:	MDC_UC_02
Use Case Name:	Login
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	This use case enables students, faculty, and admins to access the MDC Club system by entering their credentials.
Primary actor:	User (Admin / Faculty / Student)
Secondary actor:	None
Precondition:	The user should have a valid account.
Postcondition:	The system displays relevant homepage.
Main flow:	1. The user navigates to the login page. 2. The user enters their roll number/ID and password. 3. The system verifies credentials. 4. If correct, the system grants access to the appropriate dashboard (Admin, Faculty, or Student). 5. Use case ends.

EVENT CREATION & APPROVAL

Use Case Id:	MDC_UC_03
Use Case Name:	Event Creation & Approval
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	This use case allows admins to propose new events, which require approval from faculty before publishing to students.
Primary actor:	Admin
Secondary actor:	None
Precondition:	The admin should be logged into their account.
Postcondition:	Event proposal is approved or rejected by the faculty.
Main flow:	1. Admin navigates to event proposal form. 2. Admin submits event details (name, description, date, etc.). 3. Faculty reviews the submitted proposal. 4. Faculty either approves or rejects the event. 5. If approved, the event is published for students to register. 6. Use case ends.

EVENT PARTICIPATION

Use Case Id:	MDC_UC_04
Use Case Name:	Event Participation
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Students can view events and register to participate. Admins and faculty can monitor participation.
Primary actor:	Student
Secondary actor:	Admin / Faculty
Precondition:	The student must be logged into their account and have selected items to purchase.
Postcondition:	Registration status is stored in the system.
Main flow:	1. Student navigates to events page. 2. Student selects an event and registers. 3. The system saves participation details in the database. 4. Admin/Faculty can view and approve/reject participation. 5. Use case ends.

REPORT SUBMISSION & APPROVAL

Use Case Id:	MDC_UC_05
Use Case Name:	Report Submission & Approval
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Admins upload reports to the system, and faculty review, approve, or reject them.
Primary actor:	Admin
Secondary actor:	Faculty
Precondition:	The admin must be logged into their account with appropriate privileges.
Postcondition:	Report is reviewed and updated with approval/rejection status.
Main flow:	1. Admin submits a report with title, description, and file. 2. System saves the report and forwards it to the faculty. 3. Faculty reviews the report. 4. Faculty approves or rejects and provides feedback. 5. System updates the report status. 6. Use case ends.

MERCHANDISE PURCHASE

Use Case Id:	MDC_UC_06
Use Case Name:	Merchandise Purchase
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Students can browse MDC merchandise (T-shirts, etc.), place an order, and make a dummy payment.
Primary actor:	Student
Secondary actor:	Admin
Precondition:	Student must be logged in and have an active account.
Postcondition:	Order is placed and stored in the system.
Main flow:	1. Student navigates to merchandise store. 2. Student selects item, quantity, and proceeds to checkout. 3. Student selects payment method (Card/UPI Dummy). 4. System records the order and updates stock. 5. Admin can approve/reject orders. 6. Use case ends.

ATTENDANCE & CERTIFICATE GENERATION

Use Case Id:	MDC_UC_07
Use Case Name:	Attendance & Certificate Generation
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Admins take attendance for events, and students who meet the criteria can download participation certificates.
Primary actor:	Admin / Student
Secondary actor:	None
Precondition:	Student must participate in an event.
Postcondition:	Attendance is recorded, and eligible students can generate certificates.
Main flow:	1. Admin records attendance for an event. 2. Attendance is stored in the database. 3. Student logs into their dashboard. 4. If eligibility criteria are met, student can download certificate. 5. Use case ends.

MEDIA GALLERY

Use Case Id:	MDC_UC_08
Use Case Name:	Media Gallery Management
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Admin uploads photos and videos of past events, and students can view them.
Primary actor:	Admin
Secondary actor:	Student
Precondition:	Admin must be logged in.
Postcondition:	Media is visible in the student gallery.
Main flow:	1. Admin uploads photos/videos to gallery. 2. System stores media files and associates them with events. 3. Students view uploaded gallery. 4. Use case ends.

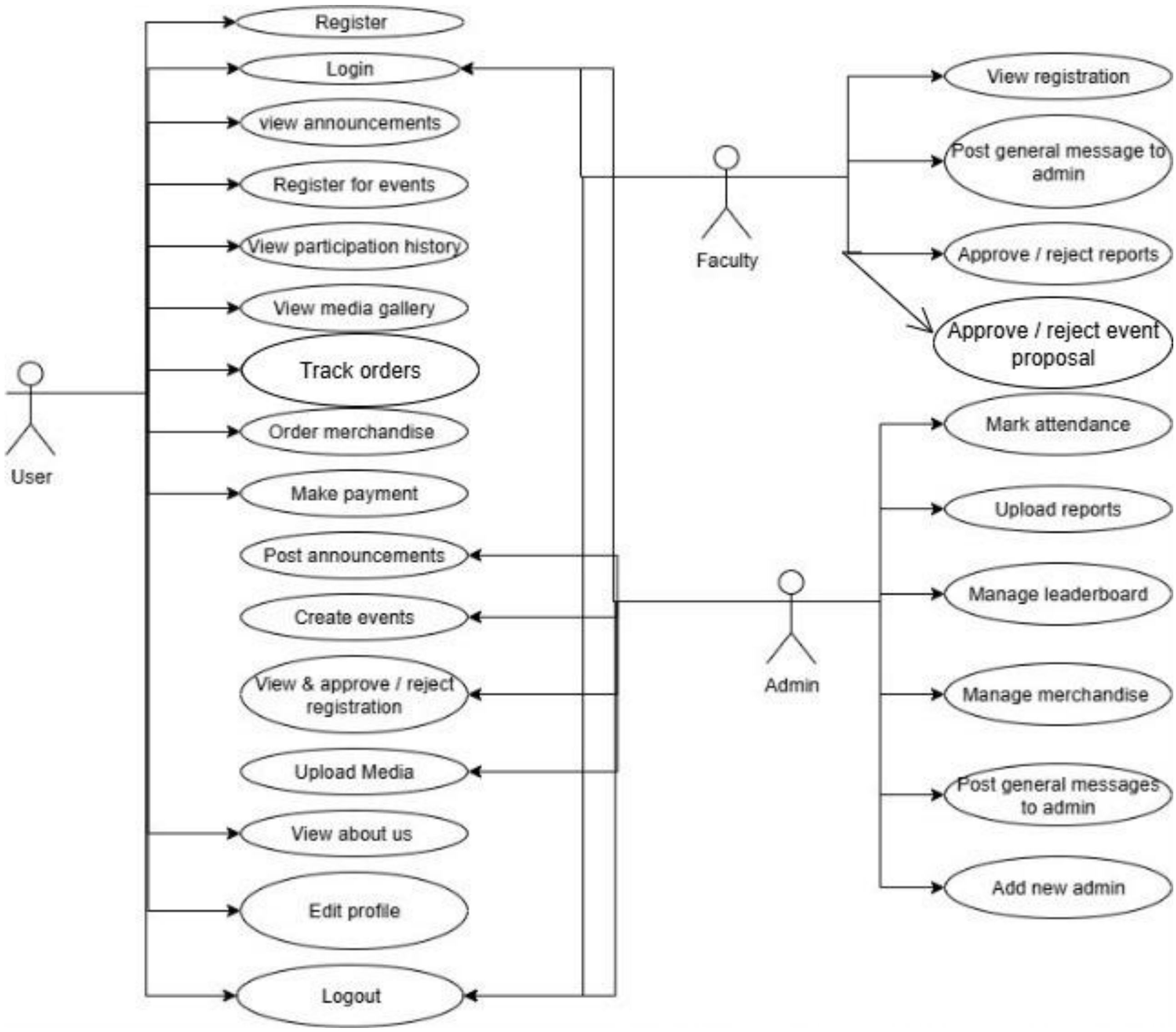
FACULTY-ADMIN CHAT

Use Case Id:	MDC_UC_09
Use Case Name:	Faculty-Admin Chat
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	Admin and faculty can exchange text messages in a chat interface for coordination.
Primary actor:	Admin/Faculty
Secondary actor:	None
Precondition:	Both must be logged into their accounts.
Postcondition:	Messages are stored in the system and displayed in real time.
Main flow:	1. Admin opens chat interface. 2. Faculty joins the chat. 3. Both exchange messages. 4. Messages are stored in database. 5. Use case ends.

LOG OUT

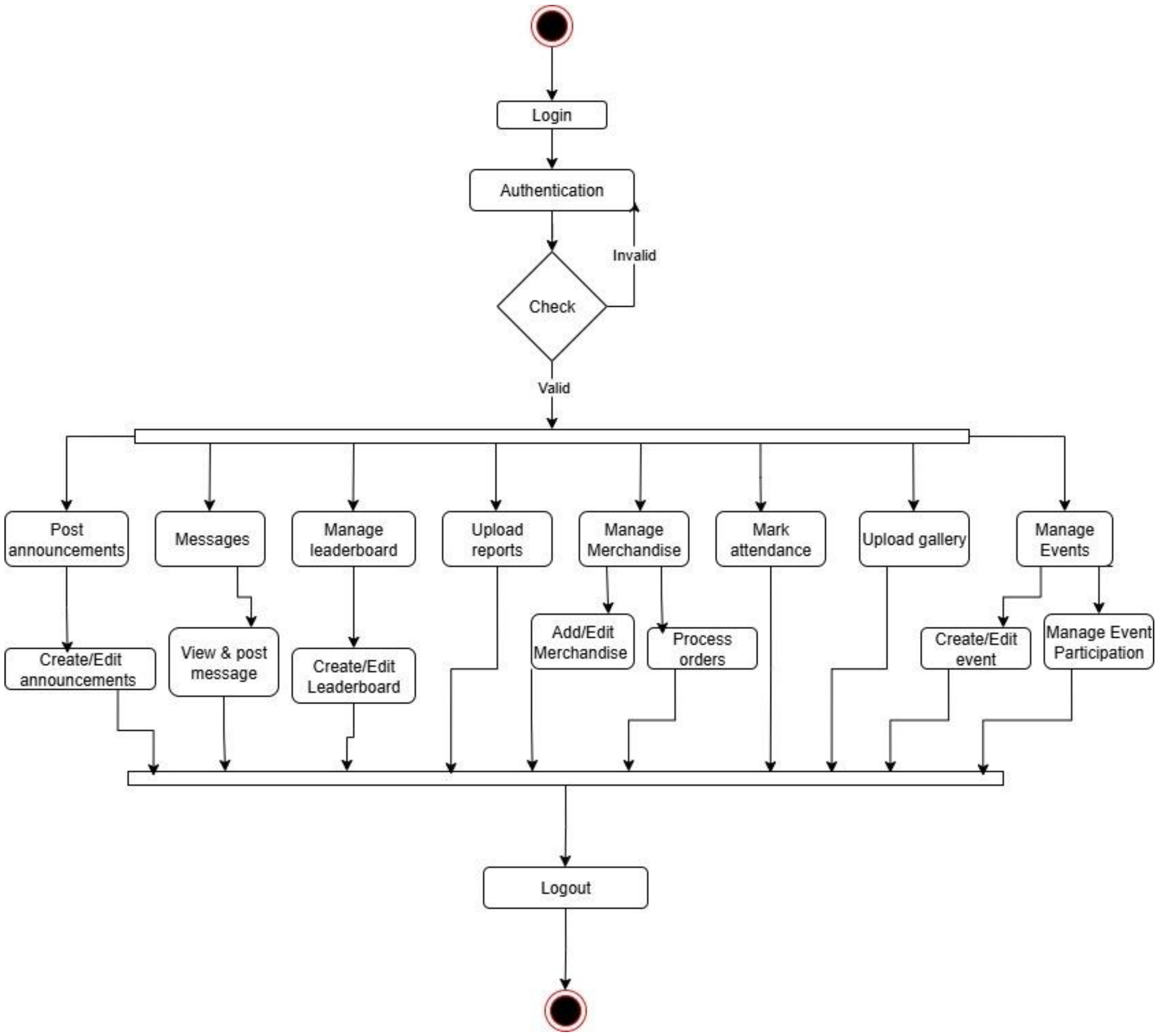
Use Case Id:	MDC_UC_10
Use Case Name:	Log Out
Created by:	Adithyan H & Melbin Antony
Date Created:	24-09-2025
Description:	This use case allows all users to securely end their session.
Primary actor:	User (Admin / Faculty / Student)
Secondary actor:	None
Precondition:	User must be logged into their account.
Postcondition:	Session ends, and user is redirected to login page.
Main flow:	1. User clicks logout button. 2. System terminates the session. 3. User is redirected to login page. 4. Use case ends.

4.3.2 USECASE DIAGRAM

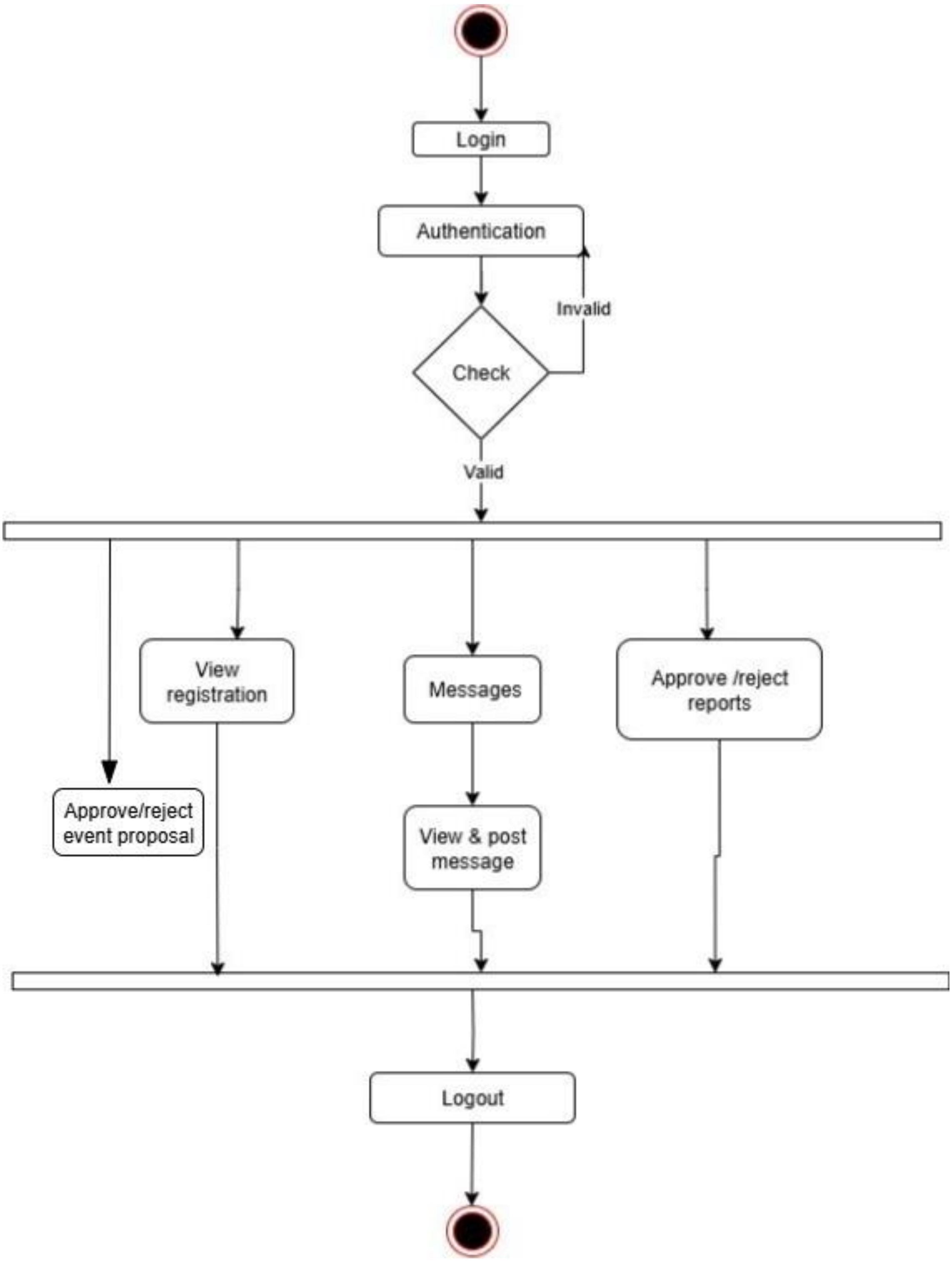


4.3.3 ACTIVITY DIAGRAMS

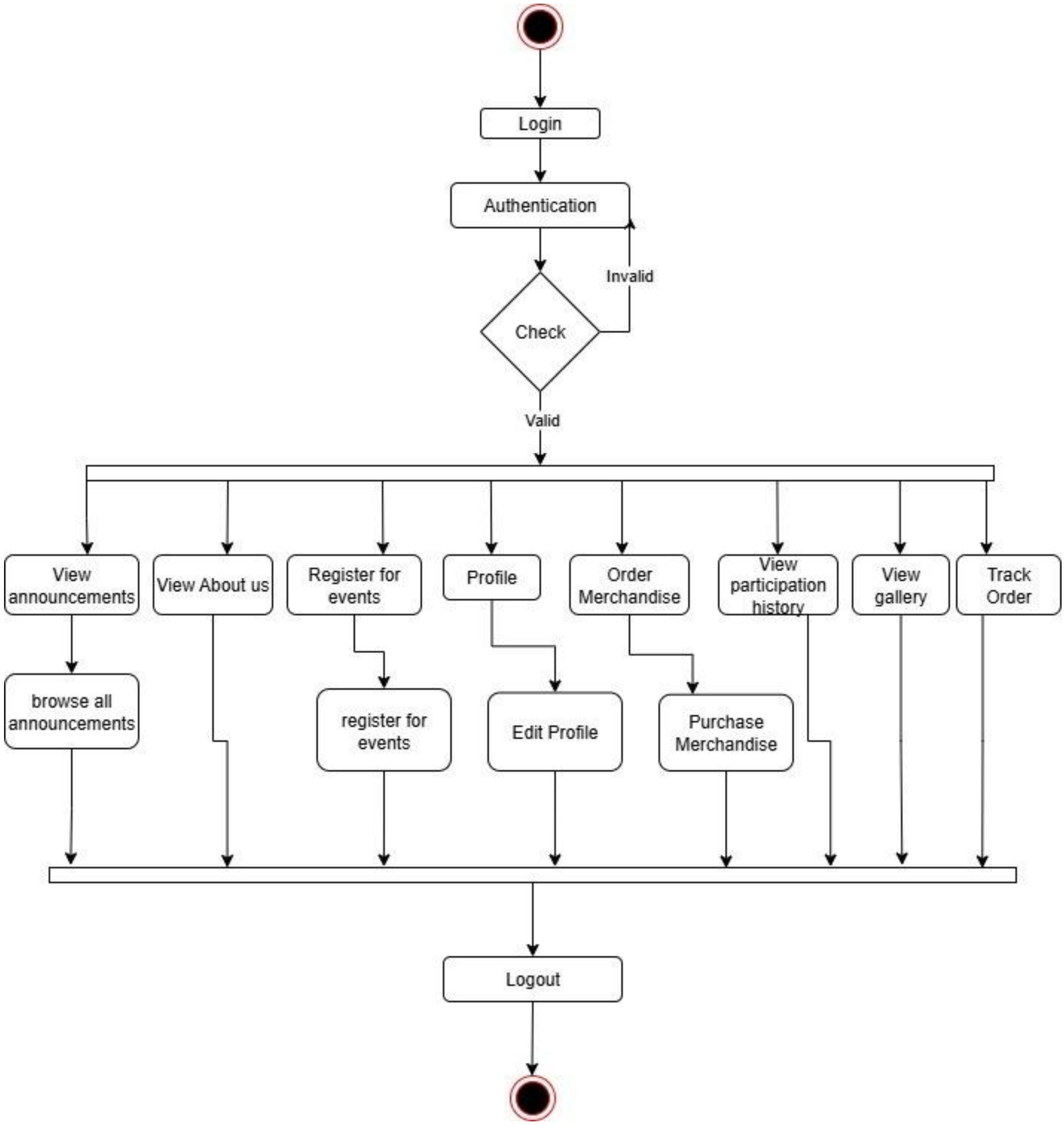
ADMIN SIDE



FACULTY SIDE



STUDENT SIDE



4.4 MODULE DETAILS

There are six main modules in the MDC Club Management System website:

- **Login Module**
- **Sign-Up Module**
- **Event Management Module**
- **Merchandise & Payment Module**
- **Attendance & Certificate Module**
- **Admin–Faculty Communication Module**

1. Login Module

The login module allows registered users (students, faculty, and administrators) to securely access the MDC Club portal. It verifies credentials (roll number/ID and password) to provide role-based access. Depending on the role, users are redirected to their respective dashboards—students can register for events and purchase merchandise, faculty can approve reports and events, and admins can manage all club activities.

2. Sign Up Module

The sign-up module enables new students and faculty members to create accounts on the MDC Club website. Users provide details including roll number, name, contact number, and password during registration. After signing up, students gain access to features such as event participation, merchandise purchase, report submission, and attendance tracking, while faculty can access event and report approval sections.

3. Event Management Module

The event management module is central to the MDC Club. Admins can create new events and propose them for faculty approval. Once approved, students can view available events and register for participation. This module also tracks participation history and status, allowing transparency for both students and administrators. Faculty can approve or reject event proposals, ensuring streamlined event scheduling.

4. Merchandise & Payment Module

This module allows students to browse and purchase club merchandise, such as T-shirts. It includes stock management, order tracking, and a **dummy payment gateway** with multiple payment methods like card and UPI. Admins can manage stock, view purchase records, and approve or reject student orders. Students can also download purchase receipts and view their purchase history.

5. Attendance & Certificate Module

The attendance and certificate module enables admins to mark student attendance for events. Attendance data is used to determine eligibility for certificates. Students can check their attendance status for each event, and if eligible, download a digitally generated certificate that includes the MDC Club logo and authorized signatures. This module strengthens event accountability and recognition.

6. Admin–Faculty Communication Module

This module provides a **chat system** between admins and faculty members. It allows real-time exchange of messages, enabling faculty to give feedback on event proposals and reports. This feature enhances coordination and ensures quick decision-making for club activities.

4.5 PERFORMANCE CONSIDERATIONS

The MDC Club Management System is designed to ensure smooth operation and efficient performance even with multiple simultaneous users (students, faculty, and admins).

Hardware Requirements

- Minimum **4 GB RAM** for optimal performance.
- Compatible with **Windows 7 or higher** and equivalent Linux distributions.
- Processor: **Dual-core 2.0 GHz or higher**.
- Storage: Minimum **10 GB free disk space** to handle database, reports, media uploads, and certificates.

Software Requirements

- **XAMPP / WAMP Server** (PHP 8.x, MySQL, Apache).
- **Web Browser**: Google Chrome, Microsoft Edge, or Mozilla Firefox.
- Frontend: **HTML, CSS, JavaScript**.
- Backend: **PHP** with **MySQL Database**.

Performance Features

- The system is designed with **normalized database tables** to reduce redundancy and improve query speed.
- **Indexing on primary keys and foreign keys** ensures quick retrieval of records such as event participation, merchandise orders, and attendance.
- Optimized for **up to 500 concurrent users** without significant performance degradation.
- Lightweight CSS and JS files ensure faster loading across devices.

4.6 SECURITY CONSIDERATIONS

Security plays a vital role in the MDC Club platform, as it involves sensitive student and faculty data such as attendance, orders, and payments.

Access Control

- **Authorized Access Only**: The system is role-based, allowing access only to authenticated users (student, faculty, or admin).
- **Unique Credentials**: Students use roll number and password, while admins and faculty use assigned IDs.

Login Security

- Passwords are stored in the database using **hashing techniques** for confidentiality.

- The login process ensures validation to prevent unauthorized access.

Data Security

- Input validation is implemented across all forms to prevent **SQL injection** and **cross-site scripting (XSS)**.
- File uploads (reports, proofs, media) are restricted to specific formats and size limits.

Transaction Security

- Although the system uses a **dummy payment gateway**, secure form handling ensures transaction details are not misused.
- Session management prevents unauthorized actions like order approvals and attendance marking.

Backup & Recovery

- Regular **database backup mechanisms** are planned to recover data in case of hardware/software failure.

4.7 TABLE DESIGN

1.Table name: announcements

Sl.no	Field Name	Data Type	Constraint	Description
1	id	int(11)	Primary Key, Not Null	Unique ID of announcement
2	message	text	Not Null	Announcement content
3	posted_on	datetime	Default current_timestamp()	Date and time of posting

2.Table name: attendance

Sl.no	Field Name	Data Type	Constraint	Description
1	attendance_id	int(11)	Primary Key, Not Null	Unique attendance record ID
2	event_id	int(11)	Foreign Key	Associated event ID
3	user_id	int(11)	Foreign Key	Associated user ID
4	status	enum('present','absent')	Not Null	Attendance status
5	date	date	Not Null	Date of event attendance

3. **Table name: contact_messages**

Sl.no	Field Name	Data Type	Constraint	Description
1	id	int(11)	Primary Key, Not Null	Unique message ID
2	name	varchar(100)	Not Null	Sender's name
3	email	varchar(100)	Not Null	Sender's email
4	message	text	Not Null	Message content
5	submitted_at	datetime	Default current_timestamp()	Time of submission

4. **Table name: delivery_confirmation**

Sl.no	**Field Name**	**Data Type**	**Constraint**	**Description**
1	**id**	**int(11)**	**Primary Key, Not Null**	**Unique delivery confirmation ID**
2	**order_id**	**int(11)**	**Foreign Key**	**Associated order ID**
3	**user_id**	**int(11)**	**Foreign Key**	**Associated user ID**
4	**delivery_notes**	**text**	**Nullable**	**Delivery notes**
5	**proof_files**	**longtext**	**JSON Valid**	**Uploaded proof file paths**
6	**submitted_at**	**timestamp**	**Default current_timestamp()**	**Submission time**

5. **Table name: events**

Sl.no	**Field Name**	**Data Type**	**Constraint**	**Description**
1	**event_id**	**int(11)**	**Primary Key, Not Null**	**Unique event ID**
2	**title**	**varchar(100)**	**Not Null**	**Title of event**
3	**description**	**text**	**Nullable**	**Event description**
4	**date**	**date**	**Not Null**	**Event date**
5	**venue**	**varchar(100)**	**Nullable**	**Venue of event**
6	**max_participants**	**int(11)**	**Nullable**	**Maximum allowed**

				participants
7	**audition_video**	**varchar(255)**	**Nullable**	**Audition video file path**

6. Table name: event_participation

Sl.no	Field Name	Data Type	Constraint	Description
1	participation_id	int(11)	Primary Key, Not Null	Unique participation ID
2	event_id	int(11)	Foreign Key	Event ID
3	user_id	int(11)	Foreign Key	User ID
4	status	enum('pending','confirmed','rejected')	Default 'pending'	Participation status
5	feedback	varchar(255)	Nullable	Feedback provided
6	audition_video	varchar(255)	Nullable	Audition video path

7. Table name: event_proposals

Sl. no	Field Name	Data Type	Constraint	Description
1	proposal_id	int(11)	Primary Key, Not Null	Proposal ID
2	admin_id	int(11)	Foreign Key	Admin ID
3	faculty_id	int(11)	Foreign Key	Faculty ID
4	title	varchar(255)	Not Null	Title of proposal
5	description	text	Nullable	Proposal description
6	proposed_date	date	Nullable	Proposed date of event
7	venue	varchar(255)	Nullable	Proposed venue
8	status	enum('pending','approved','rejected')	Default 'pending'	Approval status
9	feedback	text	Nullable	Faculty/Admin feedback
10	created_on	timestamp	Default current_timestamp()	Proposal creation time

8. **Table name: faculty_profiles**

Sl.no	Field Name	Data Type	Constraint	Description
1	user_id	int(11)	Primary Key, Not Null	Faculty user ID
2	name	varchar(100)	Not Null	Faculty name
3	phone	varchar(15)	Not Null	Faculty phone number

9. **Table name: media_gallery**

Sl.no	Field Name	Data Type	Constraint	Description
1	id	int(11)	Primary Key, Not Null	Media ID
2	event_id	int(11)	Foreign Key	Associated event ID
3	file_url	varchar(255)	Not Null	Path to media file
4	uploaded_on	datetime	Default current_timestamp()	Upload time

10. **Table name: merchandise**

Sl.no	Field Name	Data Type	Constraint	Description
1	item_id	int(11)	Primary Key, Not Null	Item ID
2	name	varchar(100)	Not Null	Item name
3	price	decimal(10,2)	Not Null	Item price
4	stock	int(11)	Default 0	Stock available
5	image_path	varchar(255)	Nullable	Item image path

11. **Table name: messages**

Sl.no	Field Name	Data Type	Constraint	Description
1	message_id	int(11)	Primary Key, Not Null	Message ID
2	sender_id	int(11)	Foreign Key	Sender user ID
3	receiver_role	enum('student','faculty','admin')	Not Null	Receiver role
4	message	text	Not Null	Message

				content
5	sent_on	datetime	Default current_timestamp()	Sent timestamp

12. **Table name: orders**

Sl. no	Field Name	Data Type	Constraint	Description
1	order_id	int(11)	Primary Key, Not Null	Order ID
2	user_id	int(11)	Foreign Key	User ID placing order
3	item_id	int(11)	Foreign Key	Merchandise item ID
4	quantity	int(11)	Not Null	Quantity ordered
5	total_price	decimal(10,2)	Nullable	Total order price
6	payment_method	varchar(50)	Not Null	Payment method
7	status	enum('pending','completed','cancelled')	Default 'pending'	Order status
8	ordered_at	datetime	Default current_timestamp()	Order placement time

13. **Table name: reports**

Sl. no	Field Name	Data Type	Constraint	Description
1	id	int(11)	Primary Key, Not Null	Report ID
2	submitted_by	varchar(100)	Not Null	Submitter
3	title	varchar(255)	Not Null	Report title
4	description	text	Nullable	Report description
5	file_path	varchar(500)	Nullable	Path to file
6	file_data	longblob	Nullable	Uploaded file content

7	**submitted_at**	**timestamp**	**Default current_timestamp()**	**Submission time**
8	**status**	**enum('submitted','pending','approved','rejected')**	**Default 'submitted'**	**Report status**
9	**feedback**	**text**	**Nullable**	**Admin/Faculty feedback**

14. Table name: student_profiles

Sl.no	Field Name	Data Type	Constraint	Description
1	user_id	int(11)	Primary Key, Not Null, Foreign Key (users.user_id)	Student unique ID
2	name	varchar(100)	Not Null	Student name
3	roll_number	varchar(50)	Unique, Not Null	Student roll number
4	department	varchar(50)	Not Null	Department name
5	phone	varchar(15)	Unique, Not Null	Student contact number

15. Table name: users

Sl.no	Field Name	Data Type	Constraint	Description
1	user_id	int(11)	Primary Key, Auto Increment, Not Null	Unique User ID
2	email	varchar(100)	Unique, Not Null	User email address
3	password	varchar(255)	Not Null	Encrypted password
4	role	enum('student','admin','faculty')	Not Null	Role of the user

CODING

5.CODING

5.1 INTRODUCTION

Coding section is where the magic happens. All the planning and the designing done in the previous sections come to life in this section. After this section can only the programmer enjoy the result of his/her hard work when he/she runs the program for the first time.

5.2 SELECTION OF SOFTWARE

PHP

PHP, an acronym for Hypertext Preprocessor, is a versatile server-side scripting language that falls under the broader category of software development. It is widely recognized for its pivotal role in web development and boasts several essential features that make it a preferred choice for building dynamic websites and web applications. Here are some of its key features:

• Open Source

• Database Integration

• Embedded in HTML

• Cross-Platform Compatibility

• Security

MYSQL

MySQL is an open-source relational database system, widely used for web development task like data storage, manipulation, and retrieval. It seamlessly integrates into web applications, eliminating the need for complex setup. MySQL is embedded within web development environments, making administrative tasks effortless. It operates as an SQL-based database, storing data in text files on the device. Unlike systems like JDBC, MySQL simplifies data access with its broad range of relational database features. Its features are

• Zero configuration

• Server less

• Stable cross platform database file

• Less memory

• Self-contained

• Transactional

5.3 CODING PHASE

The goal of the coding or programming phase is to translate the design of the system produced during the design phase into code in a given programming language, which can be executed by a computer and that performs the computation specified by the design. The coding phase affects both testing and maintenance profoundly.

The coding phase does not affect the structure of the system; it has great impact on the internal structure of modules, which affects the testability of the system. The goal of the coding phase is to produce clear simple programs. The aim is not to reduce the coding effect, but to program in a manner so that testing and maintenance costs are reduced.

Programs should not be constructed so that they are easy to write; they should be easy to read and understand. Reading programs is a much more common activity than writing programs. Hence, the goal of the coding phase is to produce simple programs that are clear to understand and modify.

5.3.1 CODING STANDARDS

The standard used in the development of the system is Microsoft Programming standards. It includes naming conversations of variables, constants and objects, standardized formats for labelling and commenting code, spacing, formatting and indenting.

Naming Conventions

The controls are prefixed to indicate their functions. The frames are prefixed with frm, textboxes are prefixed with txt, command buttons with cmd, label boxes with lbl, list boxes with lst, comboboxes with cmb, Date Time Pickers with DTP, Grid with grid and so on.

Labels and Comments

The functions of each control are labelled clearly in the GUI. The code also includes comments so that other developers using the source code in future might understand the module functions better.

TESTING

&

IMPLEMENTATION

6. TESTING

6.1 INTRODUCTION

Software testing is a critical element of software quality assurance and represents the ultimate review of specifications design and coding. Testing presents an interesting anomaly for the software. Testing is a quality measure process, which reveals the errors in the program. During testing, the program is executed with a set of test cases and the output of the program for the test cases is evaluated to determine if the program is performing as it is expected. Testing plays a very critical role in determining the reliability and efficiency of the software and it is a very important stage in software development.

6.2 TESTING

System testing is actually a series of different tests whose primary purpose is to fully exercise the computer-based systems. Although each test has a different purpose, all work to verify that all system elements have been properly integrated and perform allocated functions.

System testing is done in order to ensure that the system developed doesn’t fail at any point. Before implementations, the system is tested with experimental data to ensure that it will meets the specified requirements, special tests data are input for processing and results examined.

6.2.1 TEST PLAN

Preparation of test data

Taking various kinds of test data does the testing. Preparation of test data plays a vital role in the system testing. After preparing, the test data the system under study is tested using that test data. While testing the system by using test data errors are again uncovered and corrected by using above testing steps and correction are also noted for future use. Two kinds of test data were collected and used:

Using live test data

Live test is those that are actually extracted from organization files. After a system is partially constructed, programmers or analyst often ask users to key in a set of data fom their normal activities. Then, the system person uses this data as a way to partially test the system. In order instance, programmers or analysts extract a set of live data from the files and have entered themselves.

Using artificial test data

Artificial test data are created solely for test purpose, since they can be generated to test all combinations of formats and values. In other words, the artificial data, which can quickly be prepared by a data generating utility program in the information system department, make possible the testing of all login and control paths through the program.

The most effective test program uses artificial test data generated by person other than those who wrote the program.

In this project invalid data was entered to test whether the program would break or not. These invalid data entries were randomly generated using random people. Many people were given the software for testing the program. They use gibberish values to test if every validation holds strong.

6.3 TESTING METHODS

Testing is generally done at two levels-testing of individual modules and testing of the entire system. During system testing, the system is used experimentally to ensure that the software does not fail that is, that it will run according to its specifications and the results examined. A limited number of uses may be allowed to use the system so analysis can see whether they use it in unforeseen ways. It is preferable to discover any surprise before the organization implements the system and depends on it.

Testing is done throughout system development at various stages. It is always a good practice to test the system at many different levels at various intervals, that is, sub systems, program modules as work progresses and finally the system as a whole. During testing the major activities are concentrated on the examination ad modification of the source code. Usually, this testing is to be performed by the person other than the person who has really coded it. This is done in order to ensure more complete and unbiased testing for making the software more reliable.

There are two types of testing:

• Black box testing

• White box testing

6.3.1 WHITE BOX TESTING

In white box testing, the internal logic of the modules is considered. Following levels of testing are performed for the developed project:

6.3.1.1 Unit Testing

This involves the tests carried out on modules programs, which make up a system. This is also called as a program testing. The units in a large system many modules at different levels are needed. Unit testing focuses on the modules, independently of one another, to locate errors. The program should be tested for correctness of logic applied and should detect errors in coding. Before proceeding one must make sure that all the programs are working independently.

6.3.2 BLACK BOX TESTING

The concept of the black box is used to represent a system who's inside workings are not available for inspection. In a black box, the test item is treated as "black", since its logic is unknown; all that is known is what goes in and what comes out, or the input and output.

6.3.2.1 System Testing

The system testing is conducted on a complete, integrated system to evaluate the system's compliance with its specified requirement. It falls within scope of black box testing so no knowledge of inner design or logic is needed. As a rule, system testing takes, as its input, all of the integrated software components that have passed integration testing and also the software system itself integrated with any applicable hardware system. The purpose of the integration testing is to detect any inconsistencies between software units.

System testing is the stage of implementation, which is aimed at ensuring that the system works accurately and efficiently before live operation commence. The logical design and the physical design should be thoroughly and continually examined on paper ensure that they will work when implemented.

6.3.2.2 Integration Testing

Integration testing is a systematic technique for constructing the program structure, while at the same time conducting tests to uncover errors associated with interfacing. This is the program is constructed and tested in small segments, which makes it easier to isolate and the following common types of integration problems may be observed:

• Version mistakes

• Data integrity violations

• Overlapping function

• Resource problems especially in memory handling

• Wrong type of parameter in function calls

6.3.2.3 Validation Testing

At the culmination of the integration testing, the software was completely assembled as a package, interfacing errors have been uncovered and corrected and a final series of software validation testing began.

In validation testing we test the system functions in a manner that can be reasonably expected by customer, the system was tested against system requirement specification. Different unusual inputs that the users may use were assumed and the outputs were verified for such unprecedented inputs. Deviation or errors discovered at this step are corrected prior to the completion of this project with the help of user by negotiating to establish a method for resolving deficiencies. Thus, the proposed system under consideration has been tested by using validation testing and found to be working satisfactorily. Validation checking is performed on the: -

Numeric Field: - The numeric field can contain only numbers from 0 to 9. An entry of any character flashes an error message. The individual modules are checked for accuracy and what it has to perform. Each module is subjected to test run along with sample data. The individually tested module are integrated into a single system.

Character Field: - This field can only contain letters from A-Z and a-z. It is useful for name, address fields and so on.

Check Null Fields: - Before entering values into the database or when updating, a validation is done to check whether any NULL fields are present.

Email Fields: - A email only field with a limit of characters. All the necessary validation checks were verified to see if invalid data ever enters into the database. Null values in fields were also treated as invalid values.

Password Fields: - - A password only field with a limit of characters. All the necessary validation checks were verified to see if invalid data ever enters into the database. Null values in fields were also treated as invalid values.

6.3.3 OUTPUT TESTING

After performing validation test, the next phase is output test of the system, since no system could be useful if it does not produce the desired output in the desired format. By consideration the format of the report/output was generated or displayed and was tested. Here output format was considered in one way: on the display screen.

6.3.4 USER ACCEPTANCE TESTING

User acceptance testing (UAT) is a critical factor for the success of the **MDC Club Management System**. It ensures that the system meets the needs and expectations of its users before final deployment. During the development process, continuous interaction with end-users—students, faculty, and admins—was maintained to incorporate feedback and make necessary adjustments.

The UAT focused on the following aspects:

- **Input Screen Design:** Ensuring that all forms for registration, event proposals, attendance marking, merchandise orders, and queries are user-friendly, intuitive, and validate data correctly.
- **Output Design:** Verifying that the system produces accurate and meaningful outputs, such as event confirmations, attendance reports, order receipts, notifications, and dashboards.

Users from all three roles were selected for testing:

- **Admin Testing:** Admins logged in using their credentials to perform tasks such as creating events, approving proposals, managing merchandise, marking attendance, and viewing reports. The testing verified that all actions correctly updated the database and reflected on the dashboards.
- **Faculty Testing:** Faculty members tested features such as reviewing event proposals, approving or rejecting student submissions, and accessing reports for verification.
- **Student Testing:** Students registered on the platform, participated in events, submitted proposals, purchased merchandise, and checked attendance and order statuses. This ensured that all functionalities were accessible and worked as expected.

The feedback from UAT was used to make minor adjustments, ensuring that the system is reliable, user-friendly, and fully meets the requirements of all stakeholders.

6.4 IMPLEMENTATION

Implementation is the stage of the project when the theoretical design is turned into a working system. The implementation stage is a system project in its own right. It includes careful planning, investigation of current system and its constraints on implementation, design of methods to achieve the changeover, training of the staff in the changeover procedure and evaluation of the changeover method.

The first task in implementation is planning deciding on the methods and time scale to be adopted. Once the planning has been completed the major effort is to ensure that the programs in the system are working properly when the user has been trained.

The complete system involving both computer and user can be executed effectively. Thus, the clear plans are prepared for the activities.

Successful implementation of the new system design is a critical phase in the system life cycle. Implementation means the process of converting a new or a revised system design into an operational one.

MAINTENANCE

&

ENHANCEMENT

7. MAINTENANCE AND ENHANCEMENT

7.1 MAINTENANCE

The **MDC Club Management System** can be updated or modified as needed. Maintenance includes all activities performed after the software is deployed to ensure the system remains operational and efficient. The maintenance process involves:

- Understanding the existing software structure and workflows
- Assessing the impact of proposed changes
- Testing modifications to ensure user satisfaction and system stability

Since the system is designed using a modular approach, it requires minimal maintenance. Most potential issues are addressed during the testing phase, and any new maintenance requirement can be resolved quickly without affecting other system components.

7.2 ENHANCEMENT

The modular architecture of the MDC Club Management System allows for easy expansion and addition of new features as student, faculty, and admin needs evolve. Future enhancements could include:

- **Event Notifications and Reminders:** Automated alerts via email or SMS for upcoming events, deadlines, and announcements.
- **Advanced Attendance Analytics:** Generate detailed reports and analytics on student participation trends for better event planning.
- **Online Feedback System:** Enable students to provide feedback on events and activities, helping improve future events.
- **Enhanced Merchandise Management:** Integration of real payment gateways, delivery tracking with status updates, and stock alerts for admins.
- **Interactive Dashboard Customization:** Personalized dashboards for students and faculty with quick access to frequently used features.
- **Integration with College Portal:** Synchronize student and faculty data with existing college systems for easier management and authentication.

These enhancements will make the system more interactive, user-friendly, and adaptable to future requirements, ensuring a seamless experience for all users and supporting the growth of MDC Club activities.

CONCLUSION

8. CONCLUSION

In today's academic and extracurricular environment, technology plays a vital role in streamlining club operations and enhancing student engagement. Traditional methods of managing student activities, events, and merchandise are often time-consuming and prone to errors. The **MDC Club Management System** provides a comprehensive digital platform that efficiently connects students, faculty, and administrators, automating key processes while ensuring transparency and accountability.

The primary goal of the system is to simplify club management by offering easy access to event registrations, attendance tracking, merchandise purchases, report submissions, and communication tools. By centralizing these activities, the platform improves coordination, reduces manual workload, and provides real-time updates for all users.

Looking ahead, the MDC Club Management System is designed to support future enhancements such as advanced analytics, personalized dashboards, and integration with other institutional systems. These improvements will further enhance user experience, increase participation, and ensure the club operates efficiently in a growing digital environment.

BIBLIOGRAPHY

BIBLIOGRAPHY

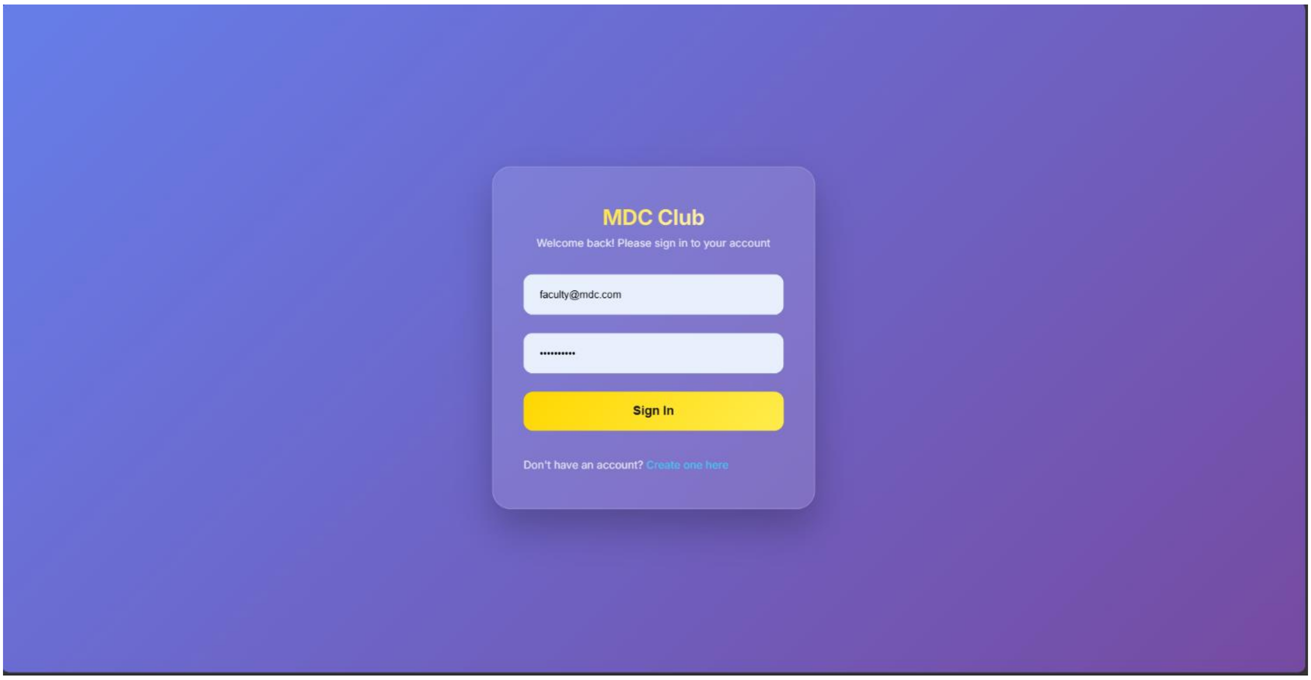
WEBSITES

- Chatgpt
- www.wikipedia.org
- www.tutorialpoint.com
- www.pinterest.com
- www.pexels.com
- www.w3schools.com

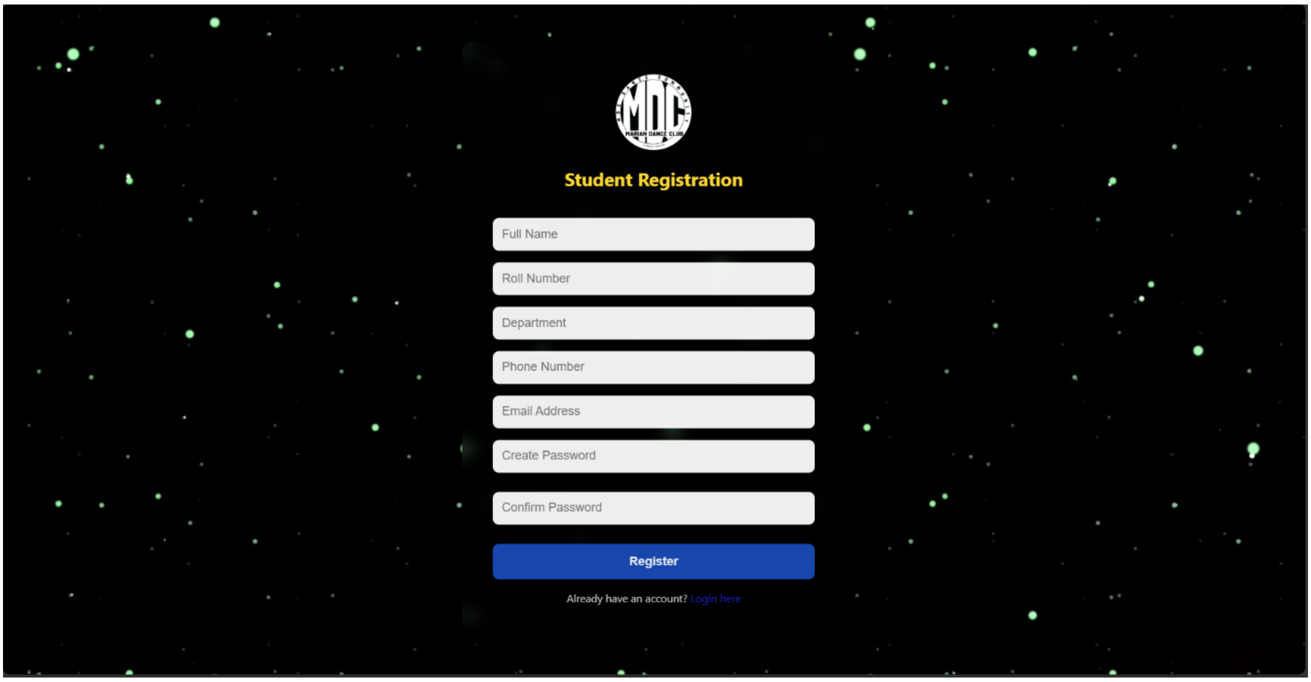
APPENDIX

SCREENSHOTS

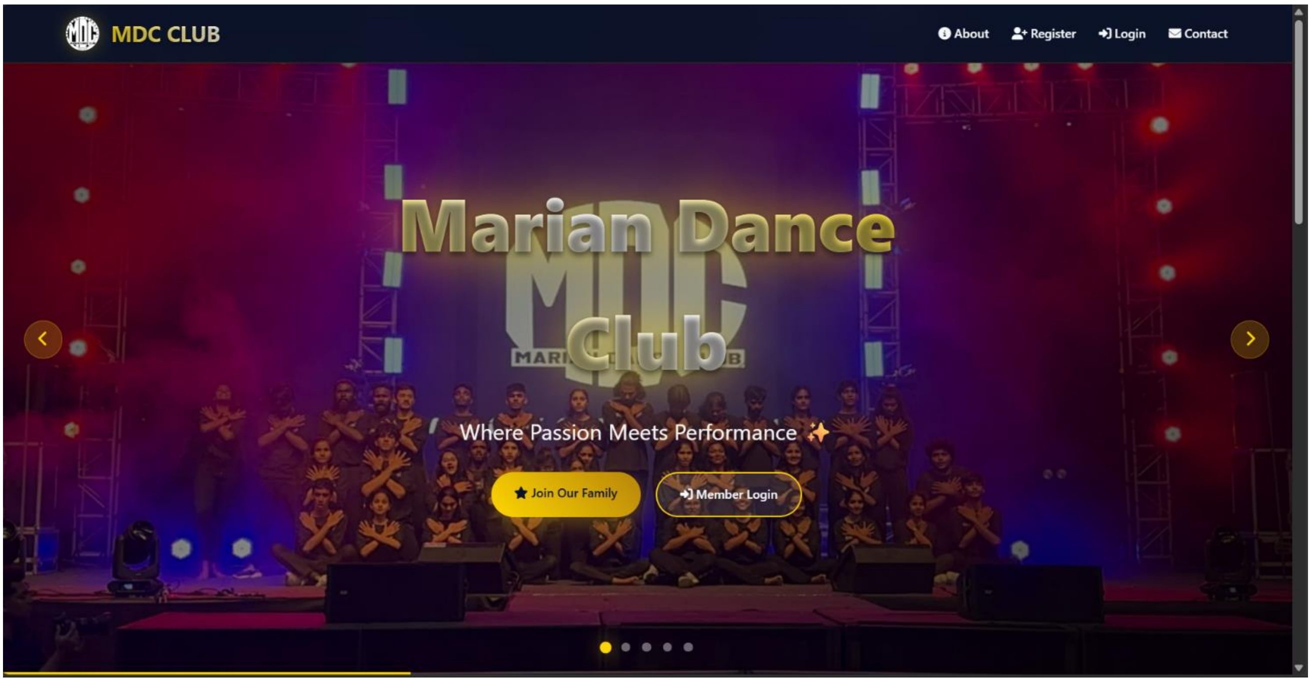
1. LOGIN PAGE



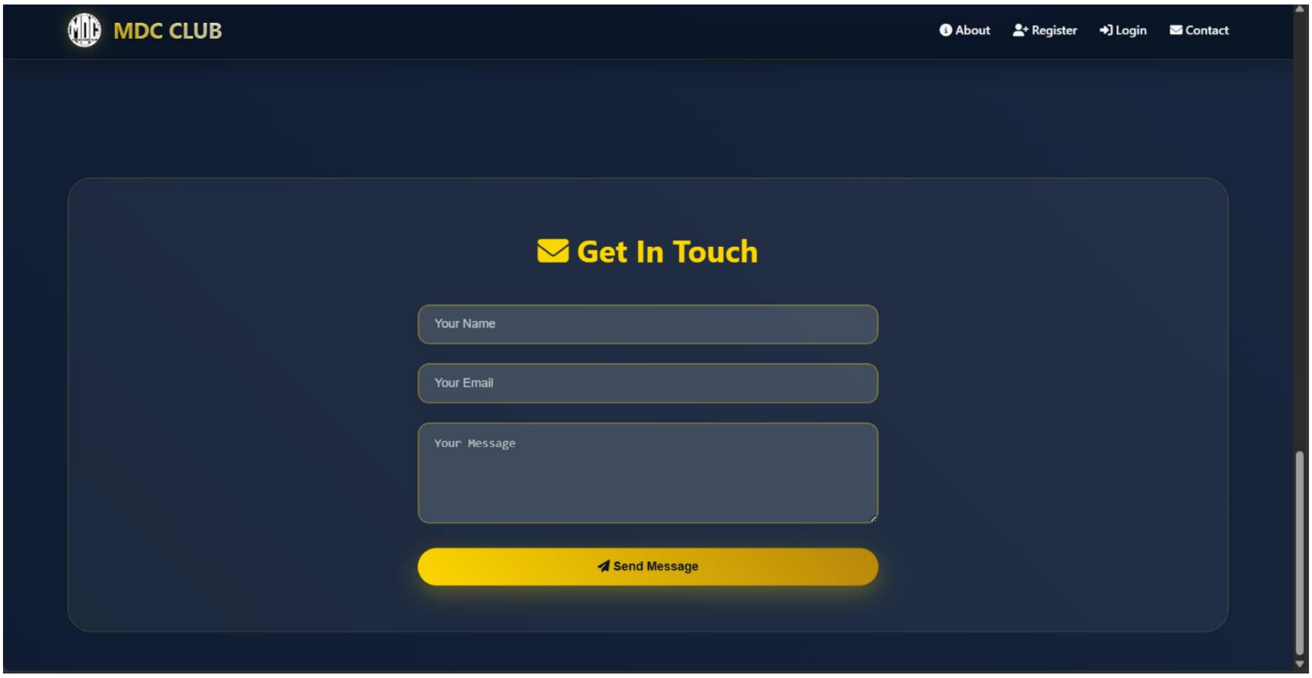
2. REGISTRATION PAGE



3. HOME PAGE

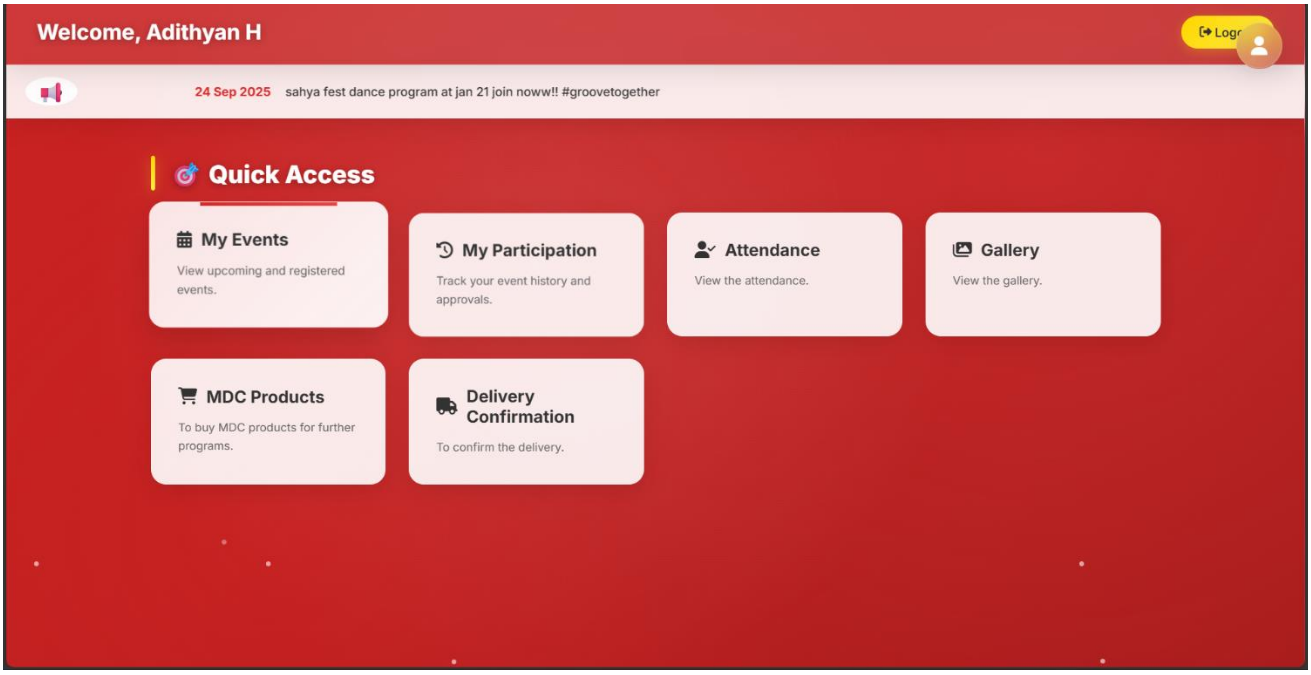

MDC CLUB
About
Register
Login
Contact
Marian Dance Club
Where Passion Meets Performance ✨
Join Our Family
Member Login


Diverse Styles
From classical to hip-hop, contemporary to bollywood - explore every dance form that moves your soul.
Competitions
Showcase your talent in inter-collegiate competitions and win hearts with your performances.
Community
Join a family of passionate dancers who support, inspire, and grow together.
About MDC
The Marian Dance Club is where dreams take flight and rhythms come alive. We're more than just a dance club - we're a vibrant community that celebrates the art of movement and expression.
From energetic flashmobs that light up the campus to mesmerizing performances at college events, MDC has been the heartbeat of Marian College's cultural scene.
Whether you're a seasoned dancer or someone who's never stepped foot on a dance floor, MDC welcomes everyone with open arms. Come, be part of our journey!

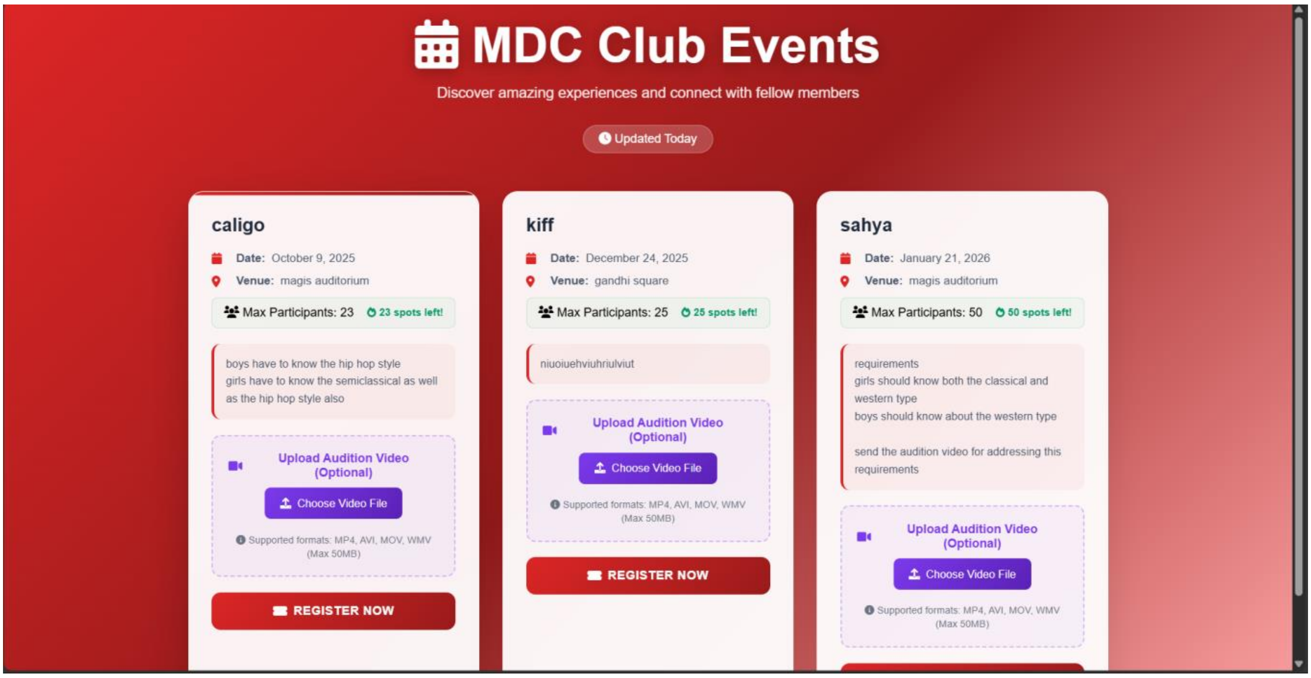

MDC CLUB
About
Register
Login
Contact
Get In Touch
Your Name
Your Email
Your Message
Send Message

STUDENT MODULE

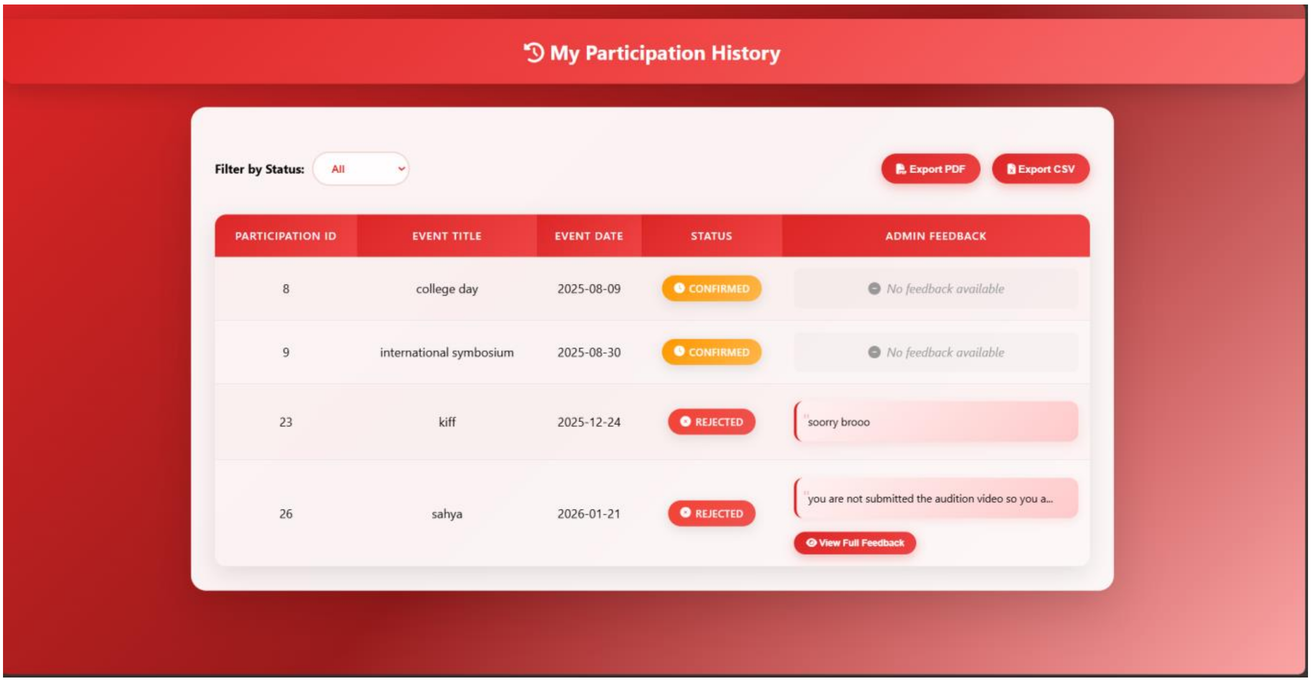
1. DASHBOARD PAGE



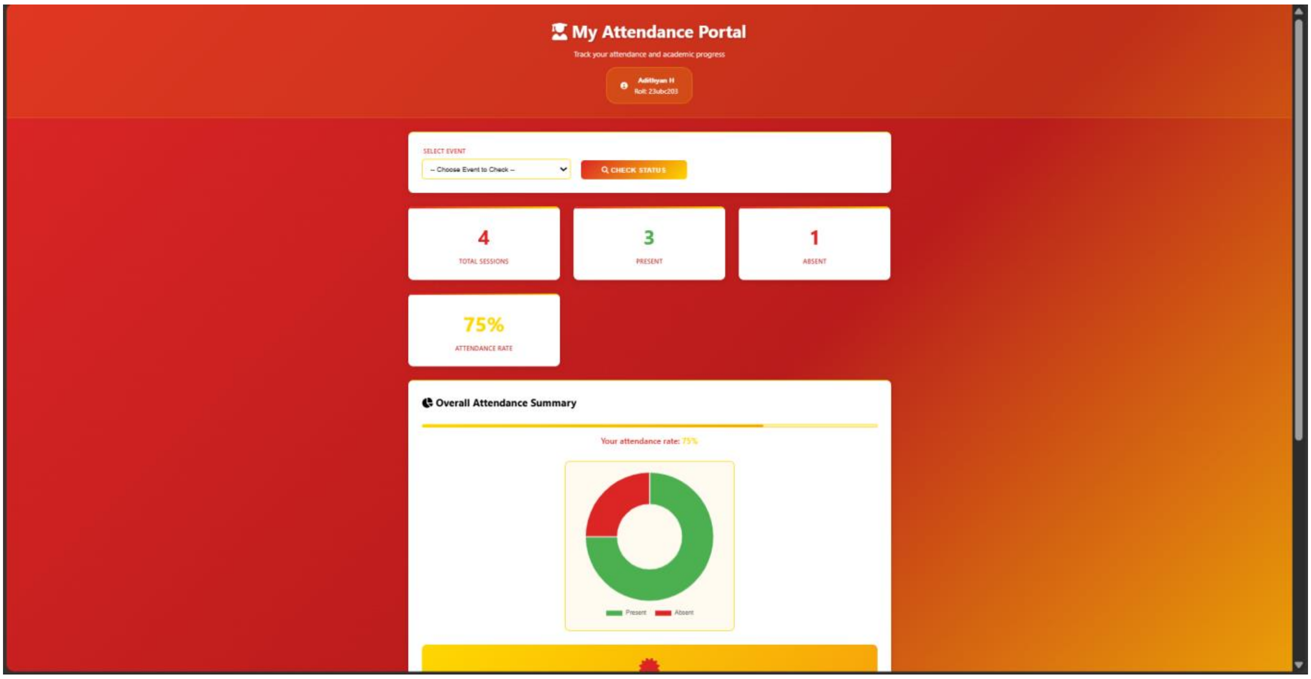
2. EVENT REGISTRATION PAGE

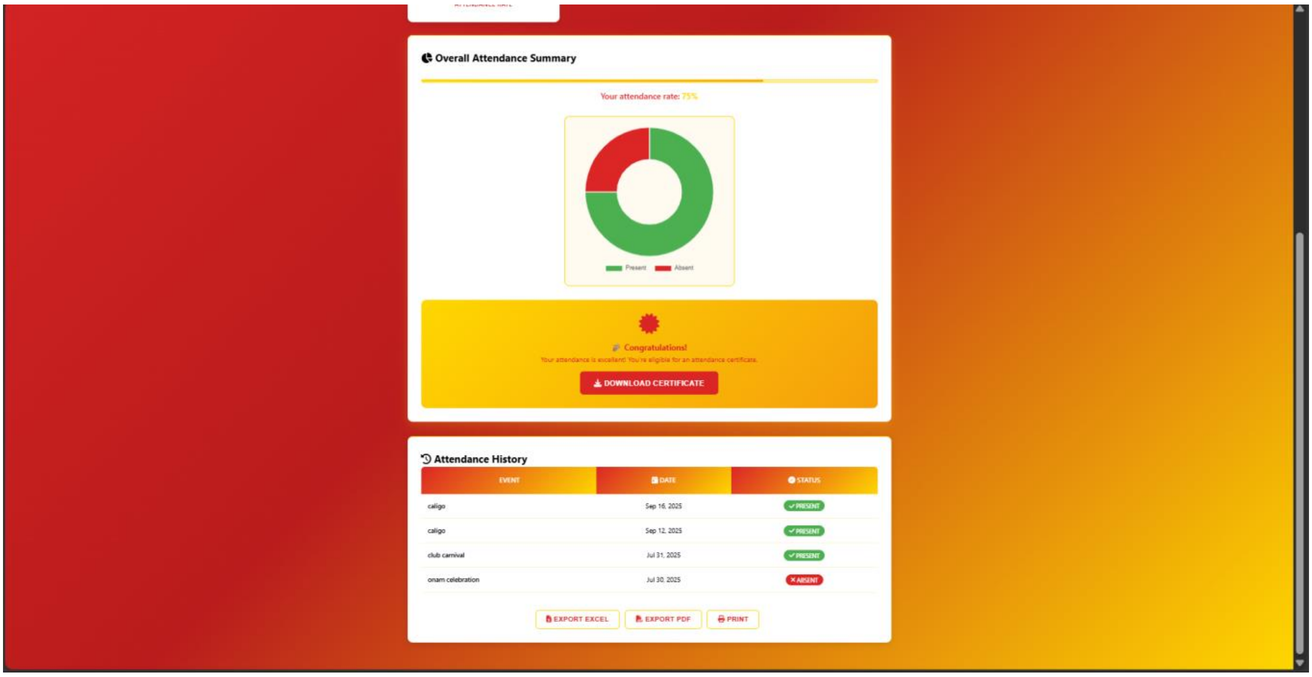


3. PARTICIPATION HISTORY PAGE

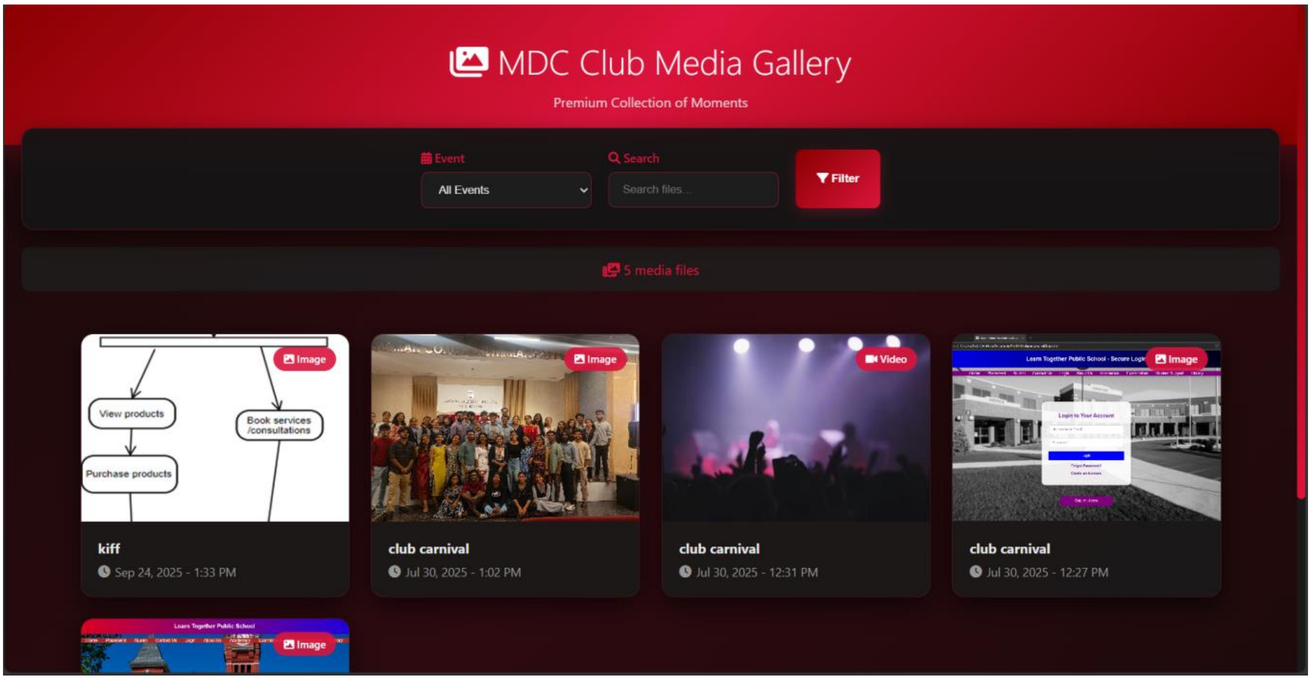


4. ATTENDANCE VIEWING PAGE

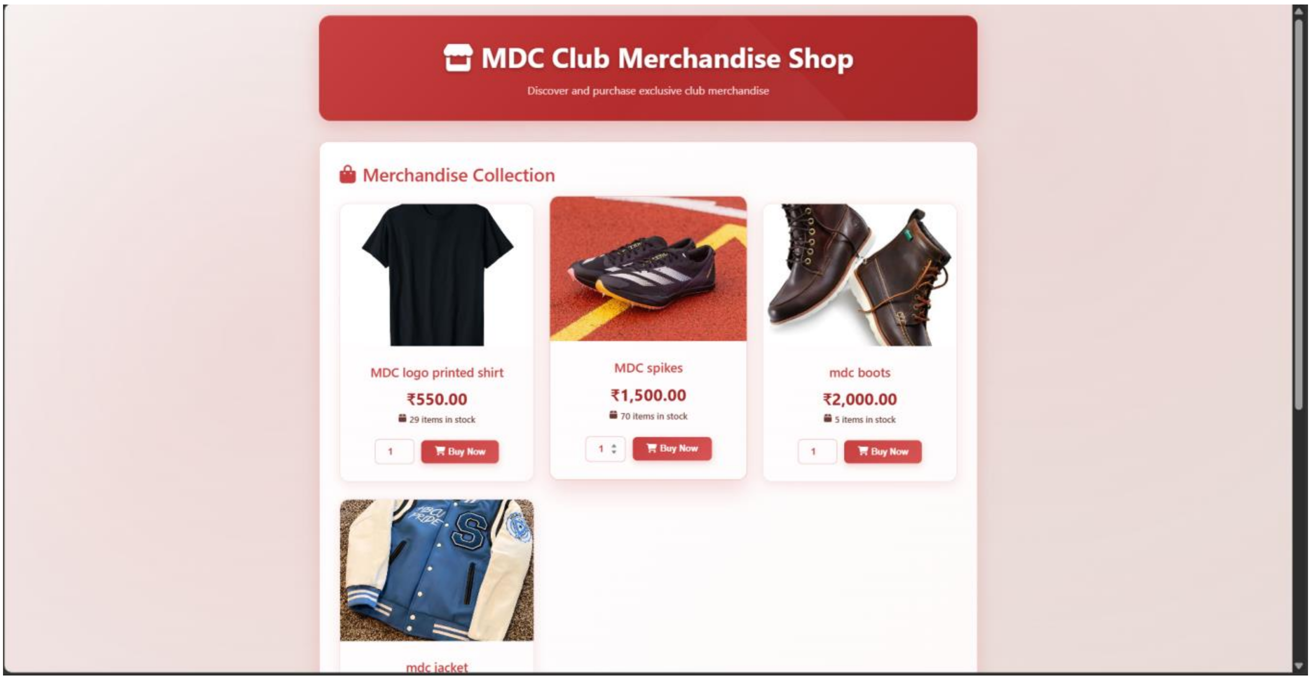


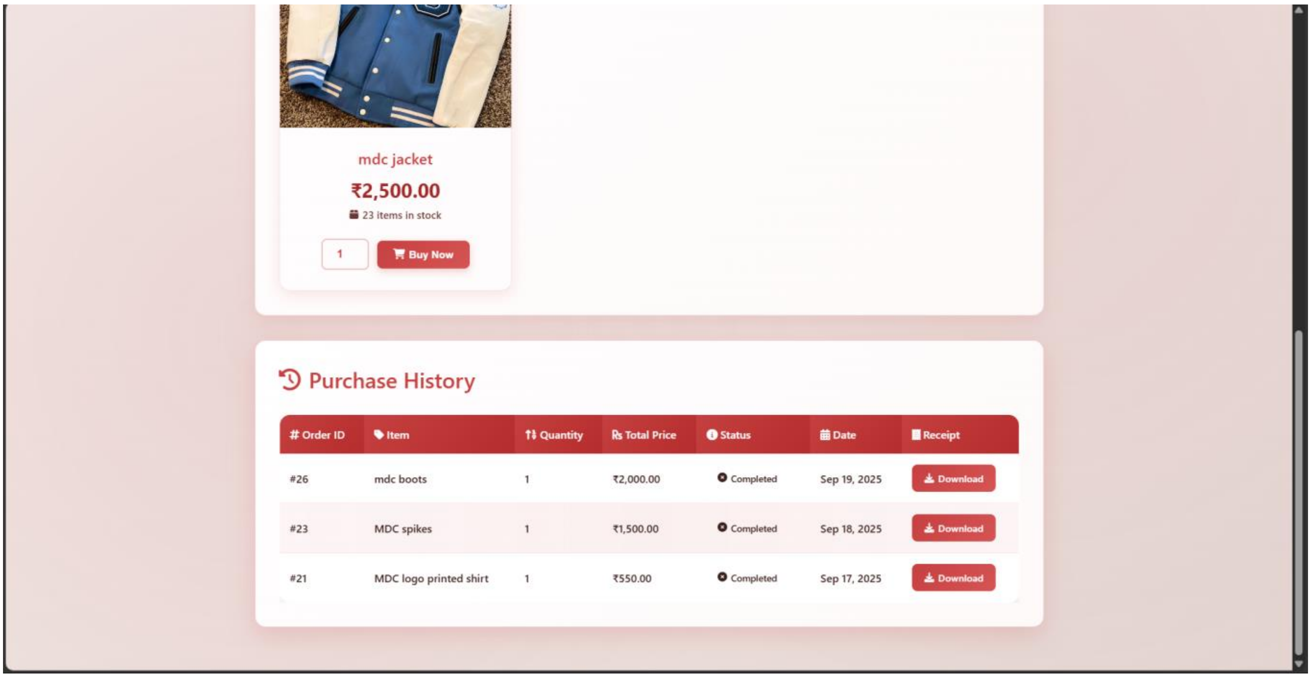


5. MEDIA VIEWING PAGE

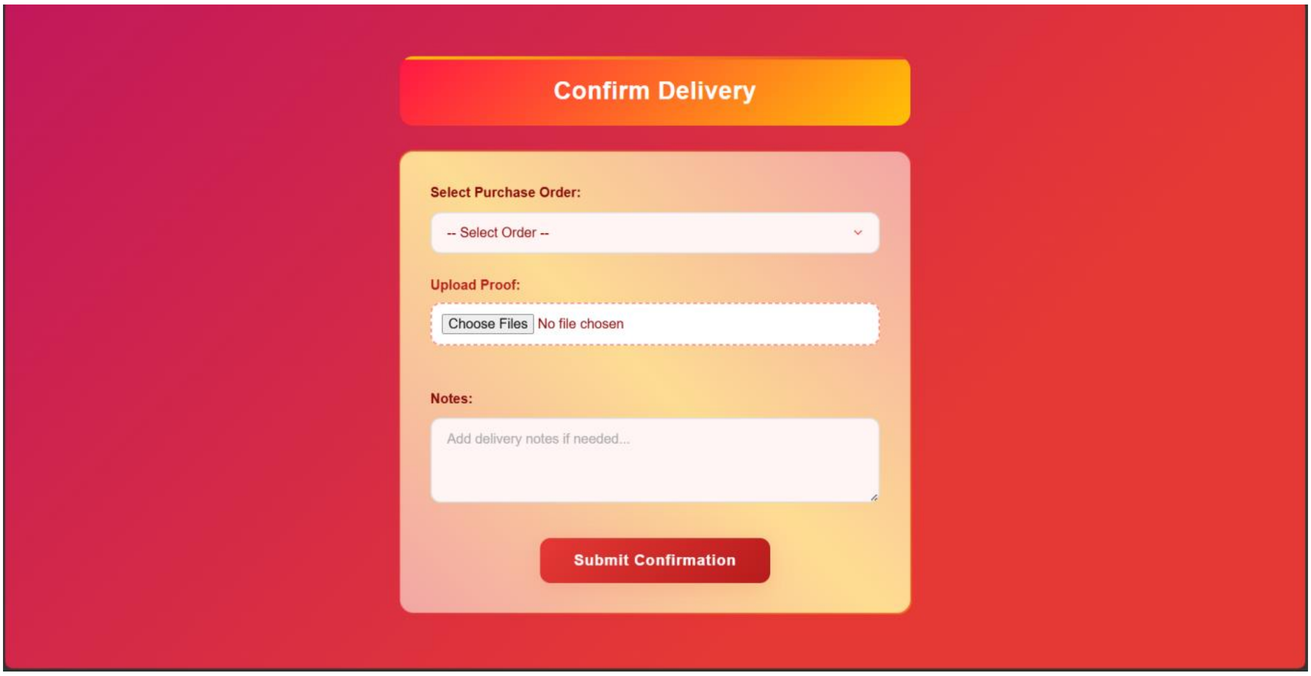


6. MERCHANDISE PAGE



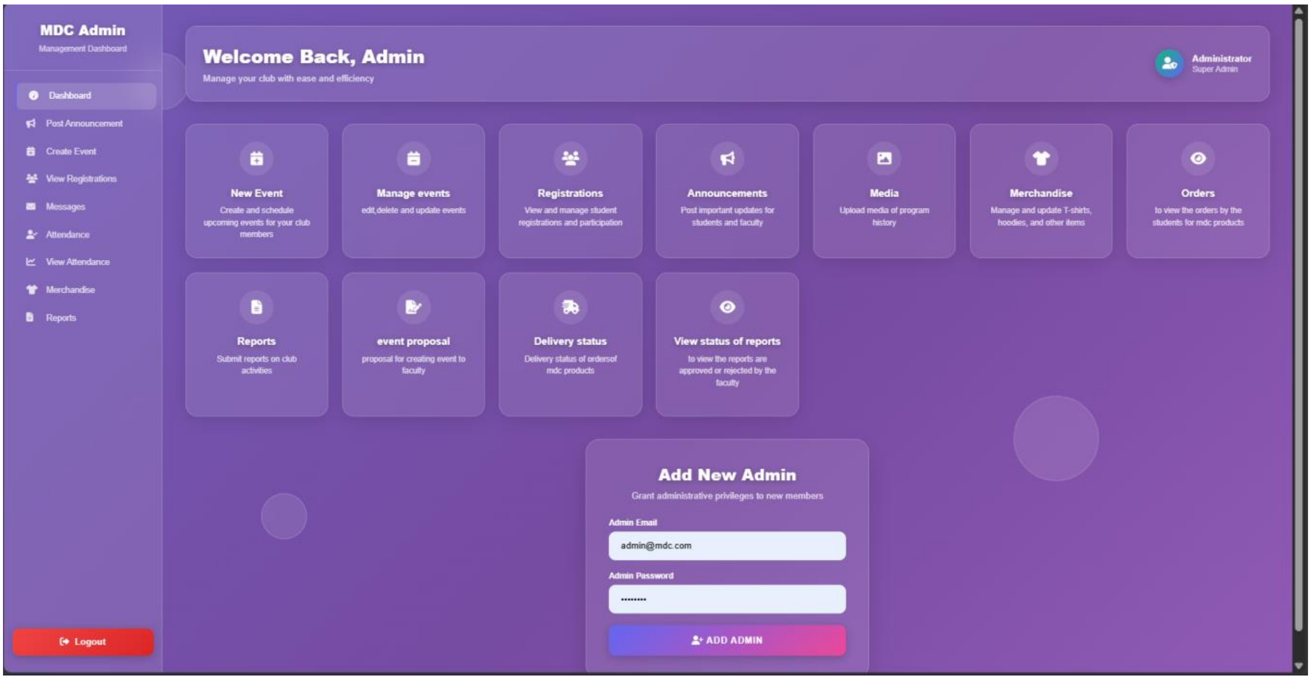


7. DELIVERY CONFIRMATION PAGE

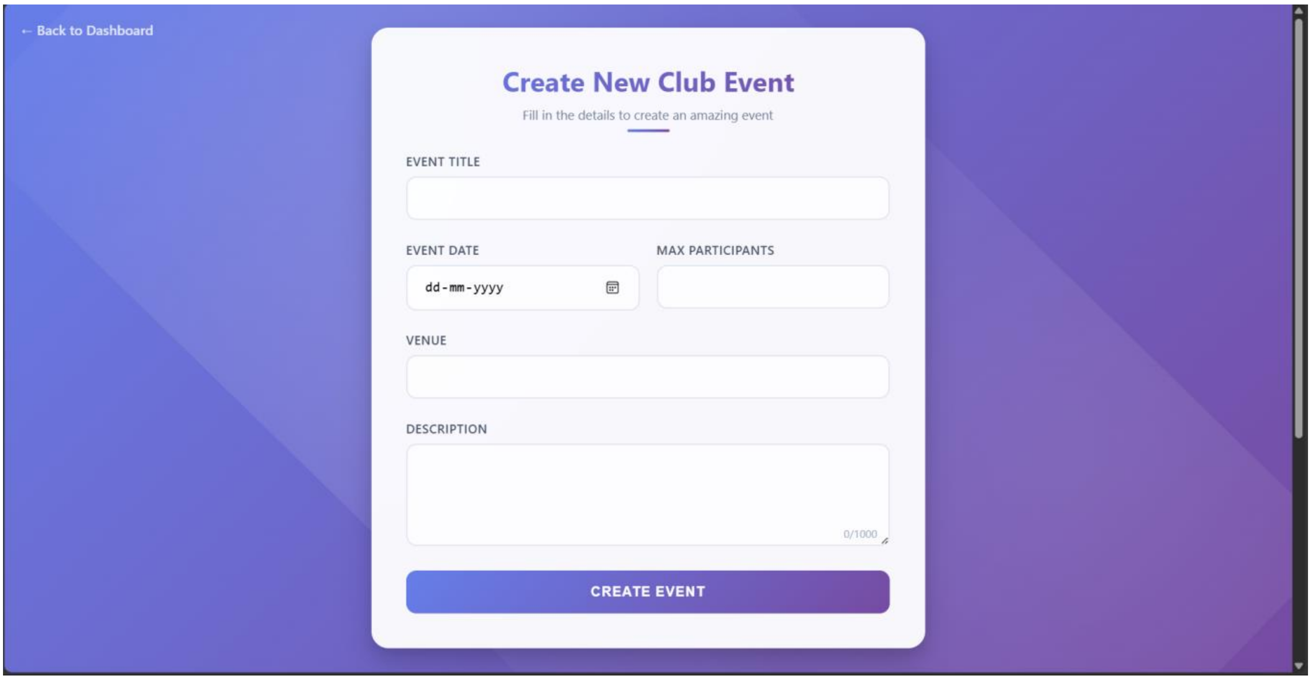


ADMIN MODULE

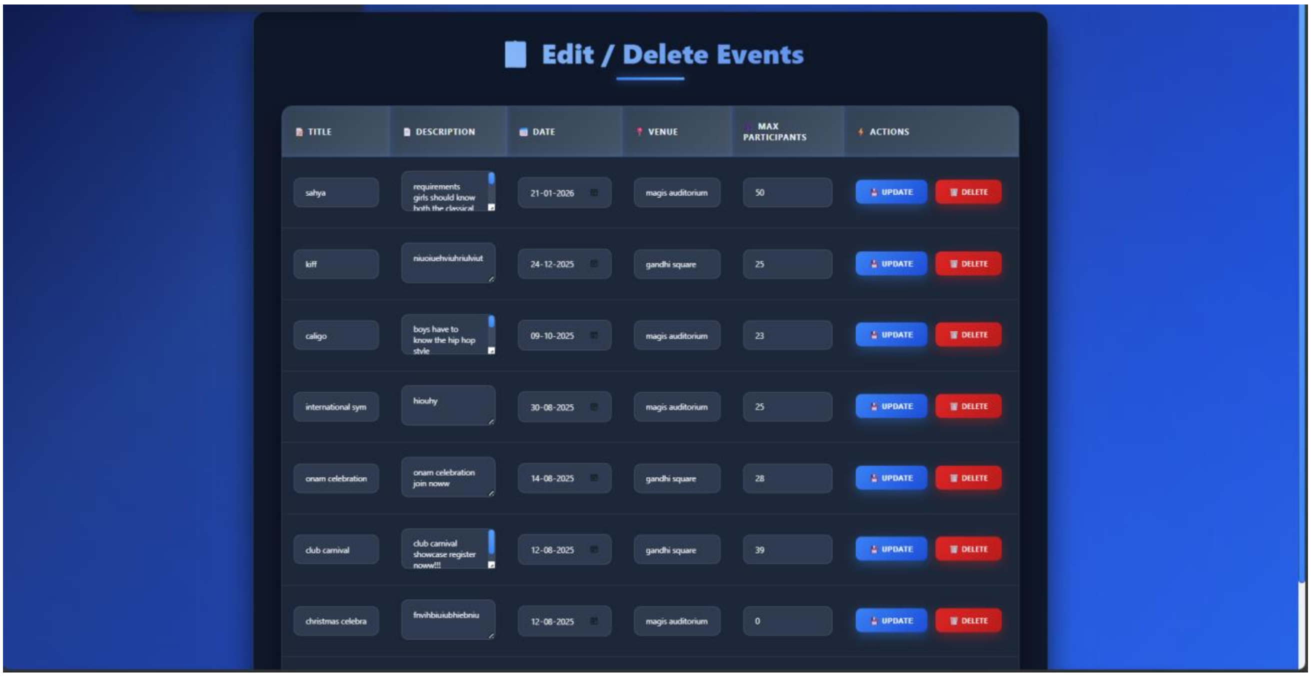
1. DASHBOARD PAGE



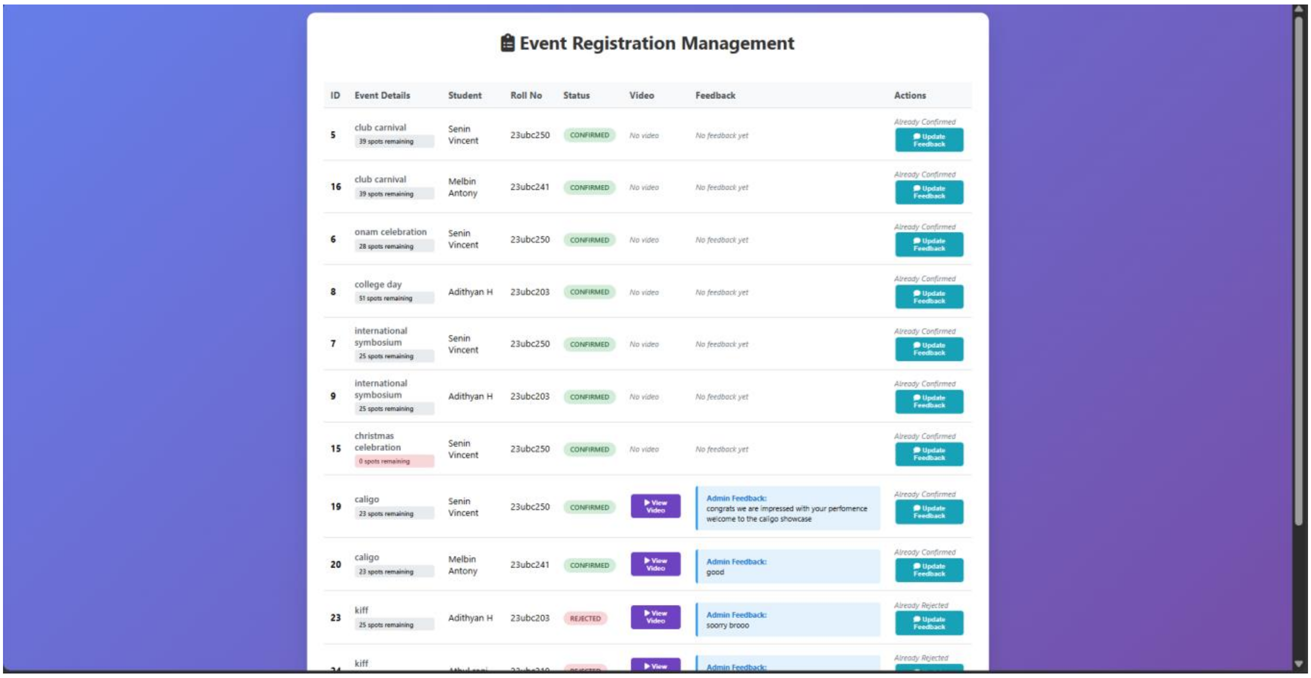
2. EVENT CREATION PAGE



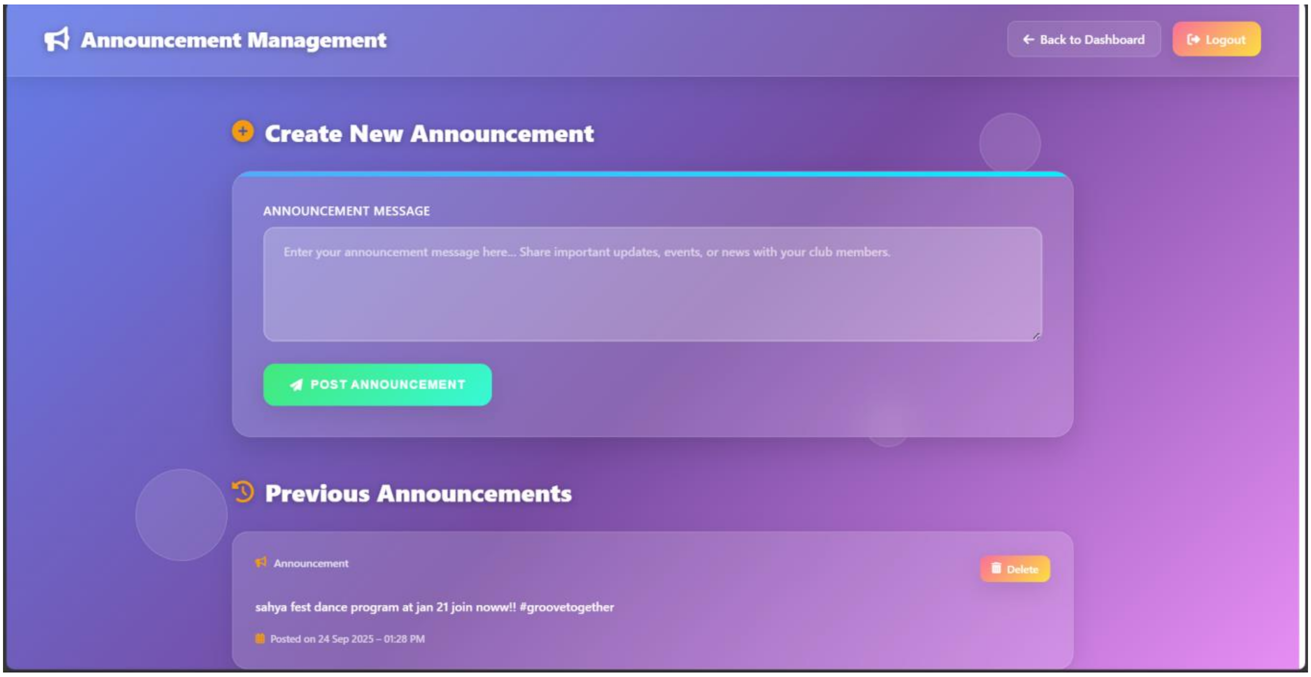
3. MANAGING EVENTS PAGE



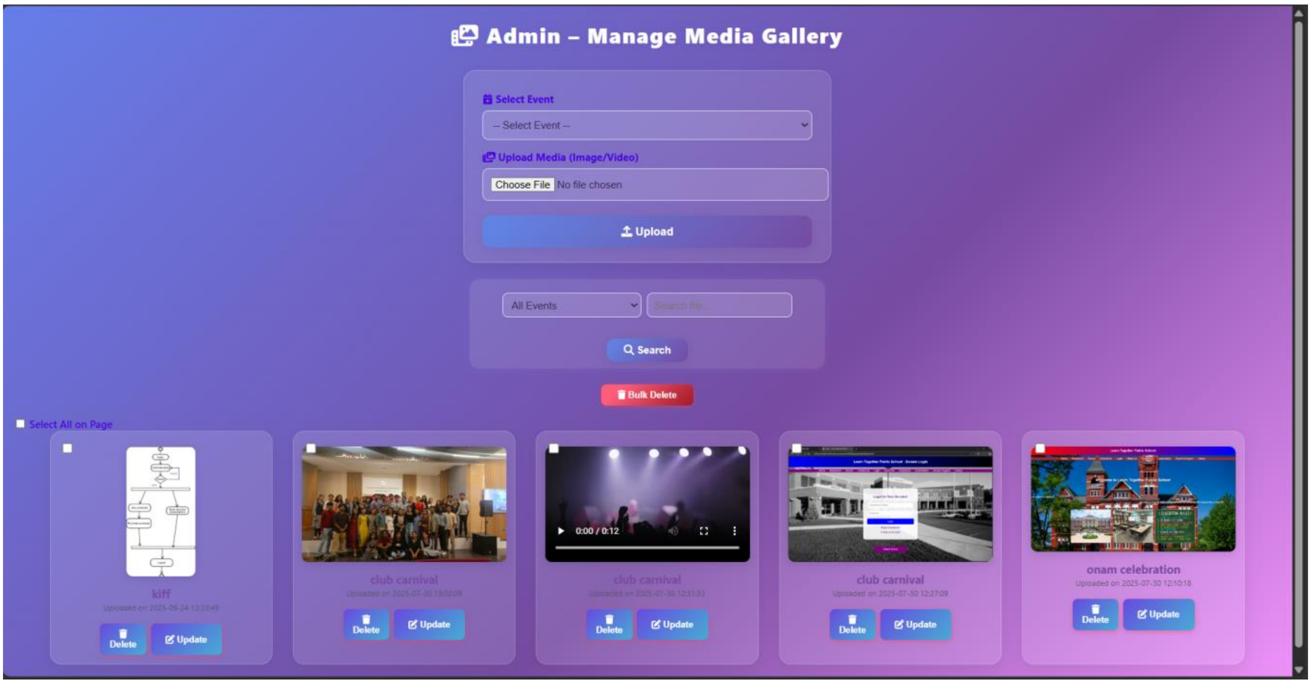
4. MANAGE REGISTRATION PAGE



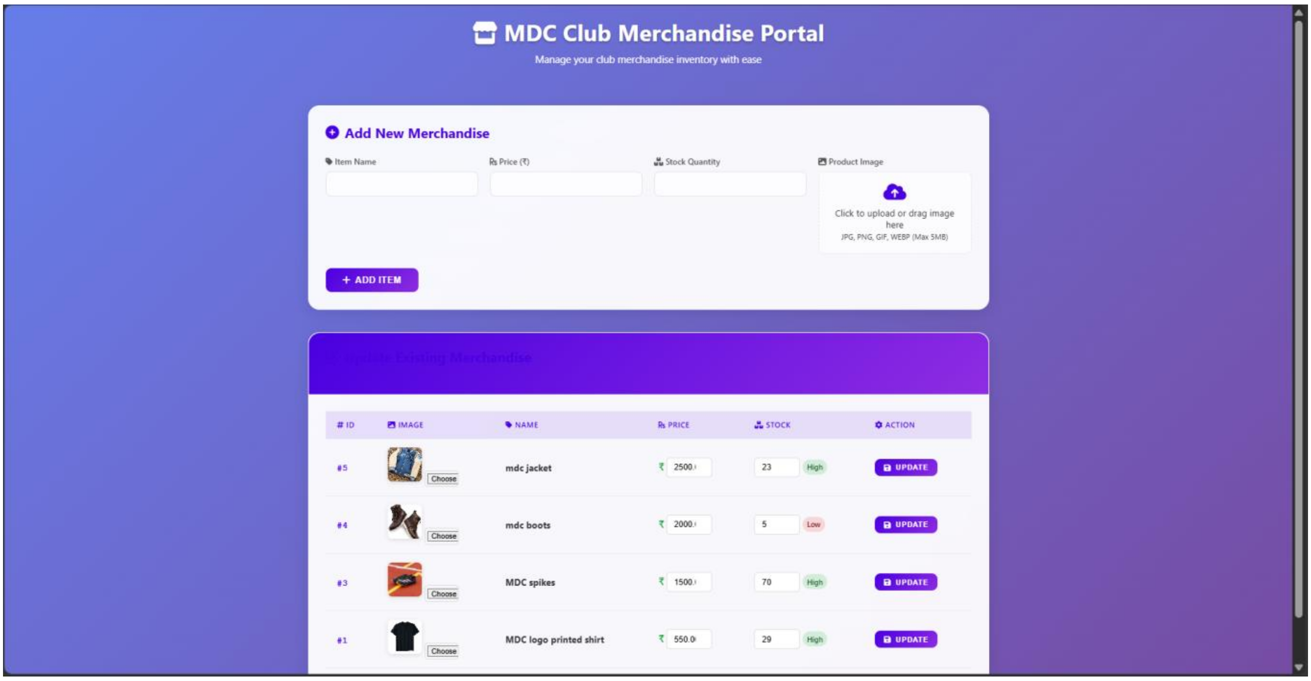
5. ANNOUNCEMENT CREATION PAGE



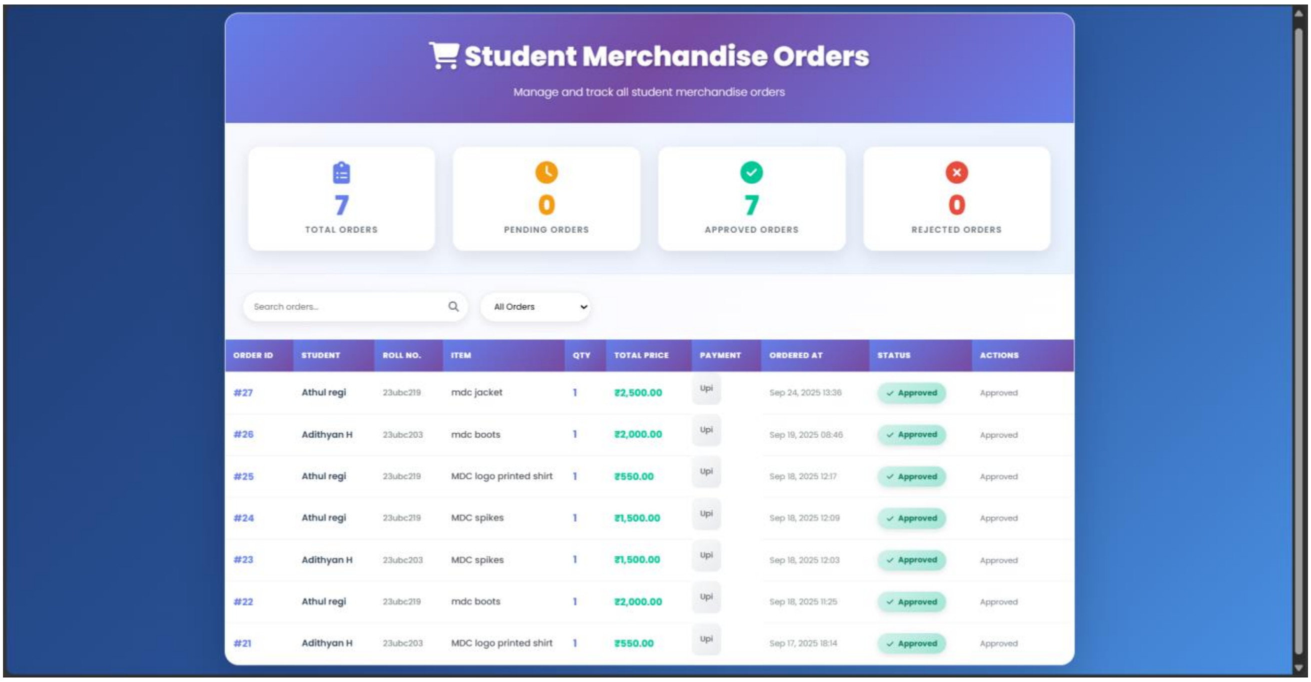
6. MANAGING MEDIA PAGE



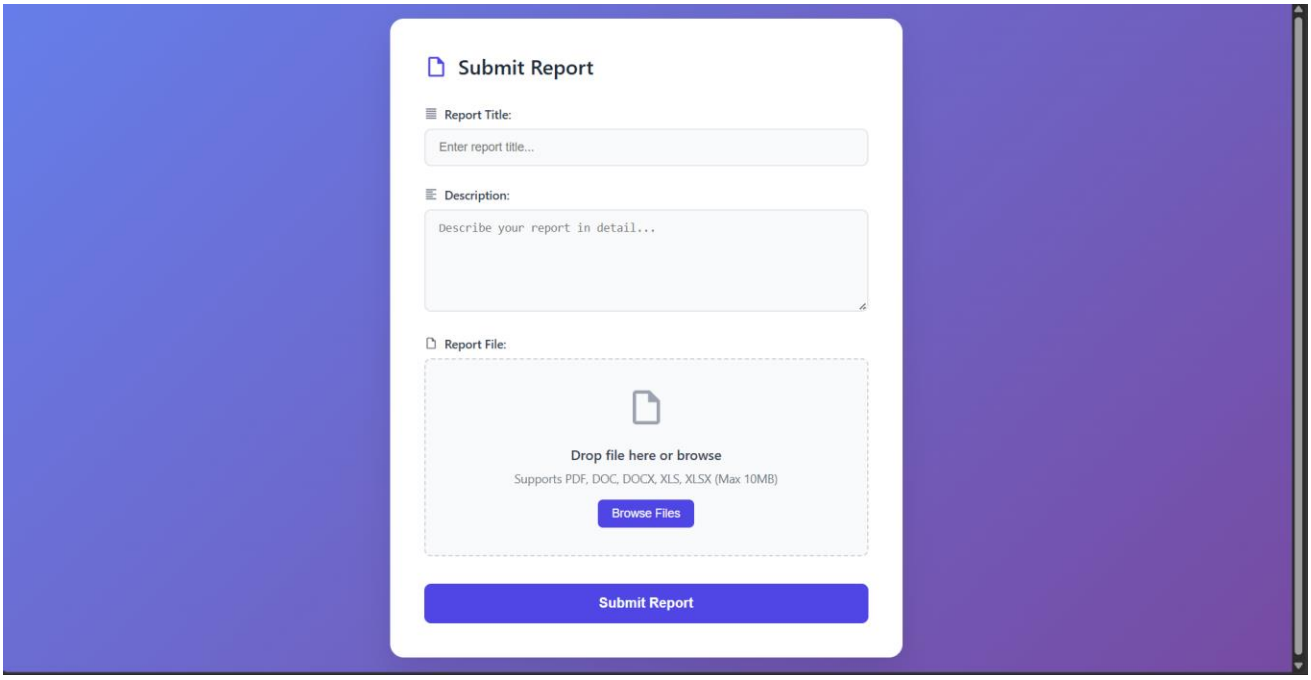
7. MANAGE MERCHANDISE PAGE



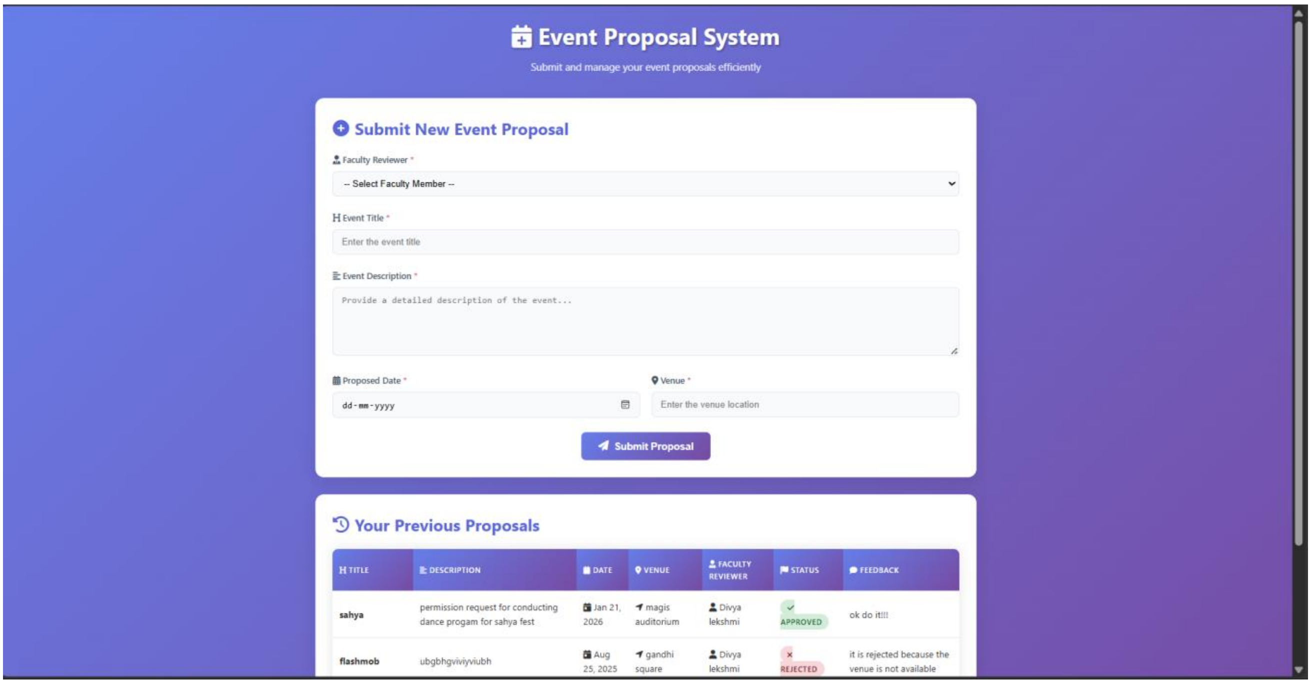
8. VIEW ORDERS PAGE



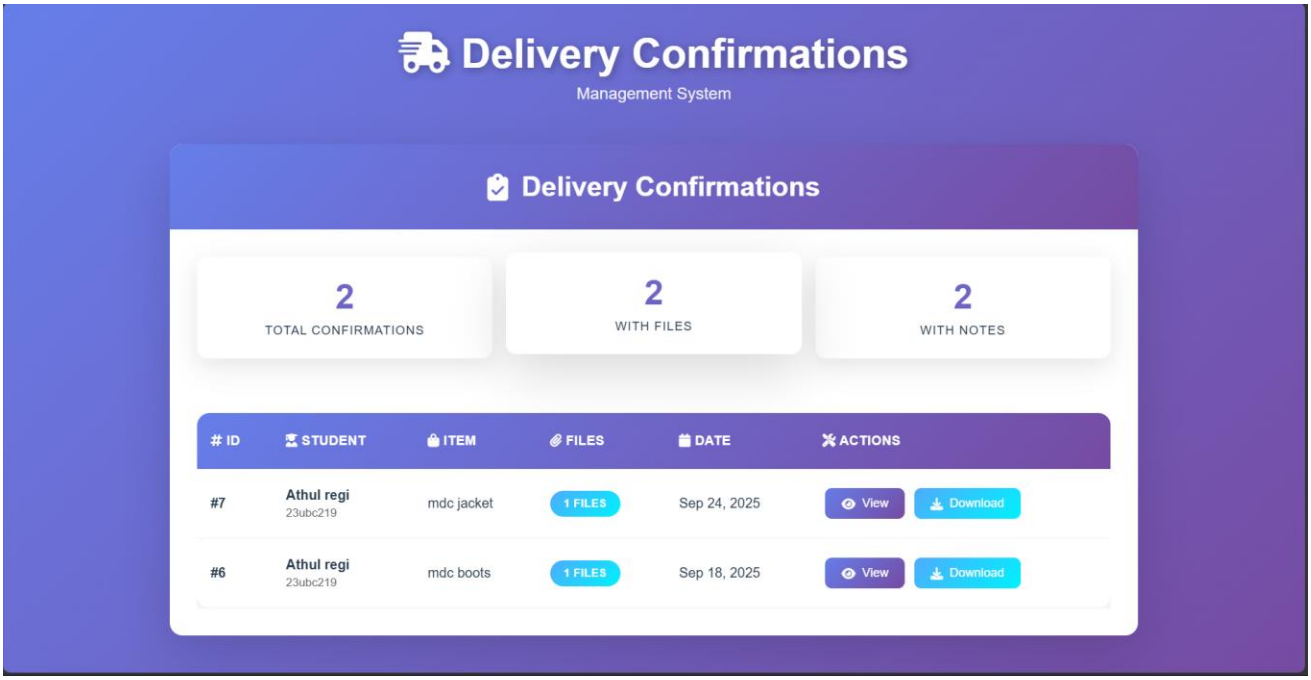
9. REPORT SUBMISSION PAGE



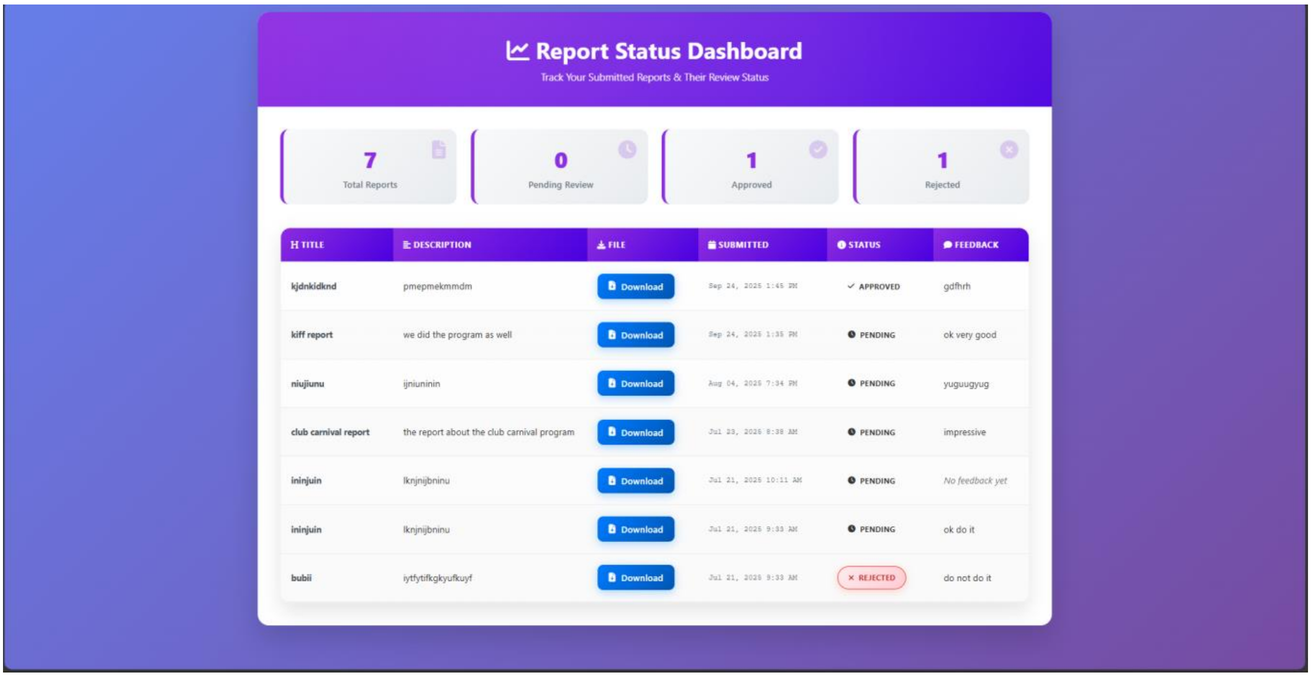
10. EVENT PROPOSAL PAGE



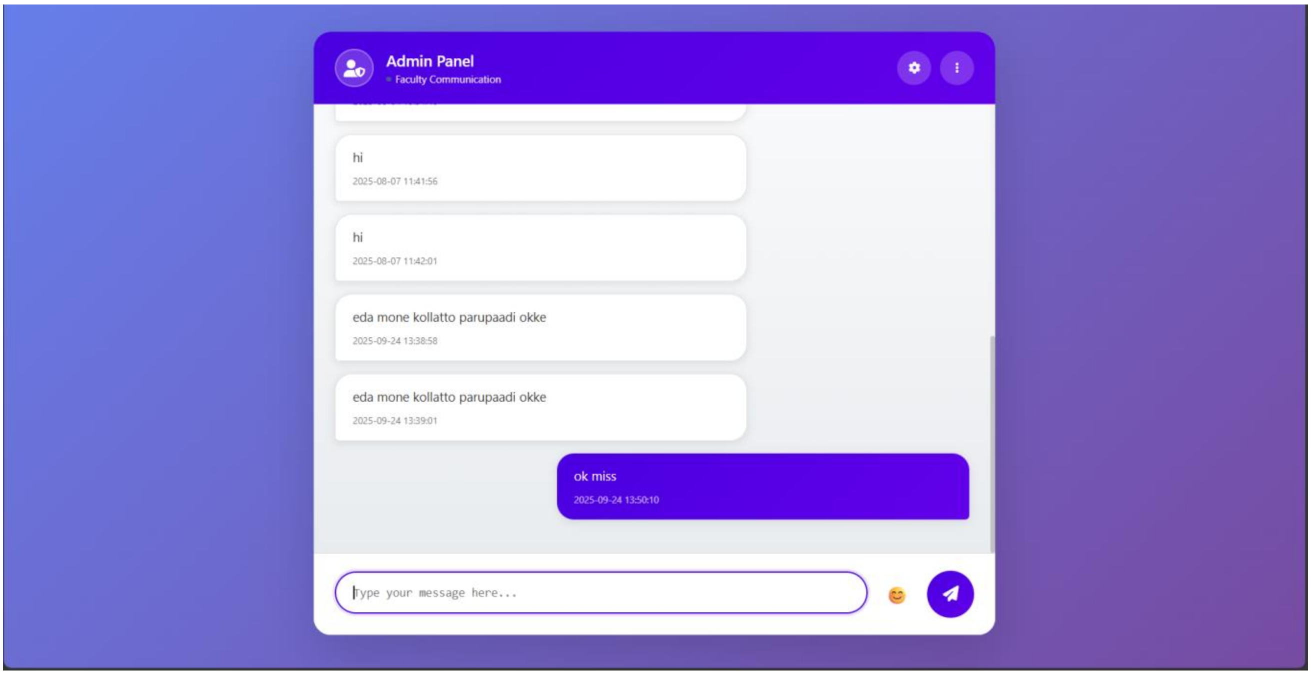
11. VIEW DELIVERY CONFIRMATION PAGE



12. REPORT STATUS VIEWING PAGE

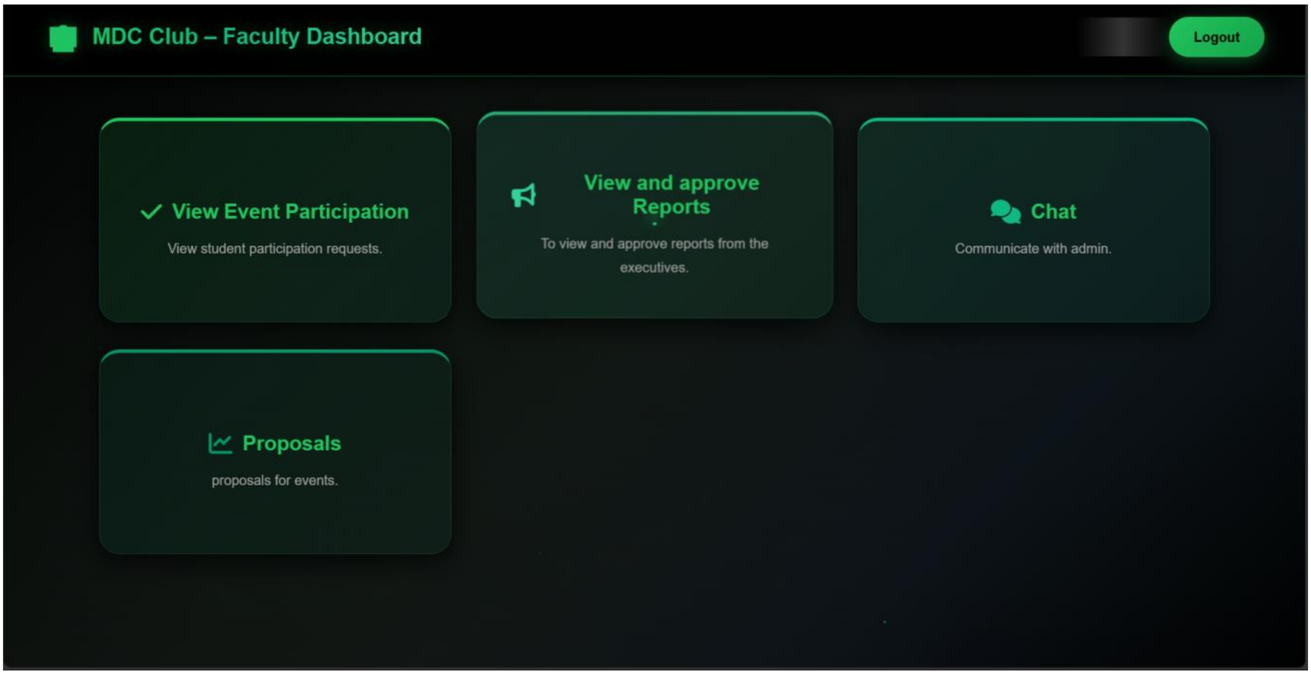


13. CHAT WITH FACULTY PAGE

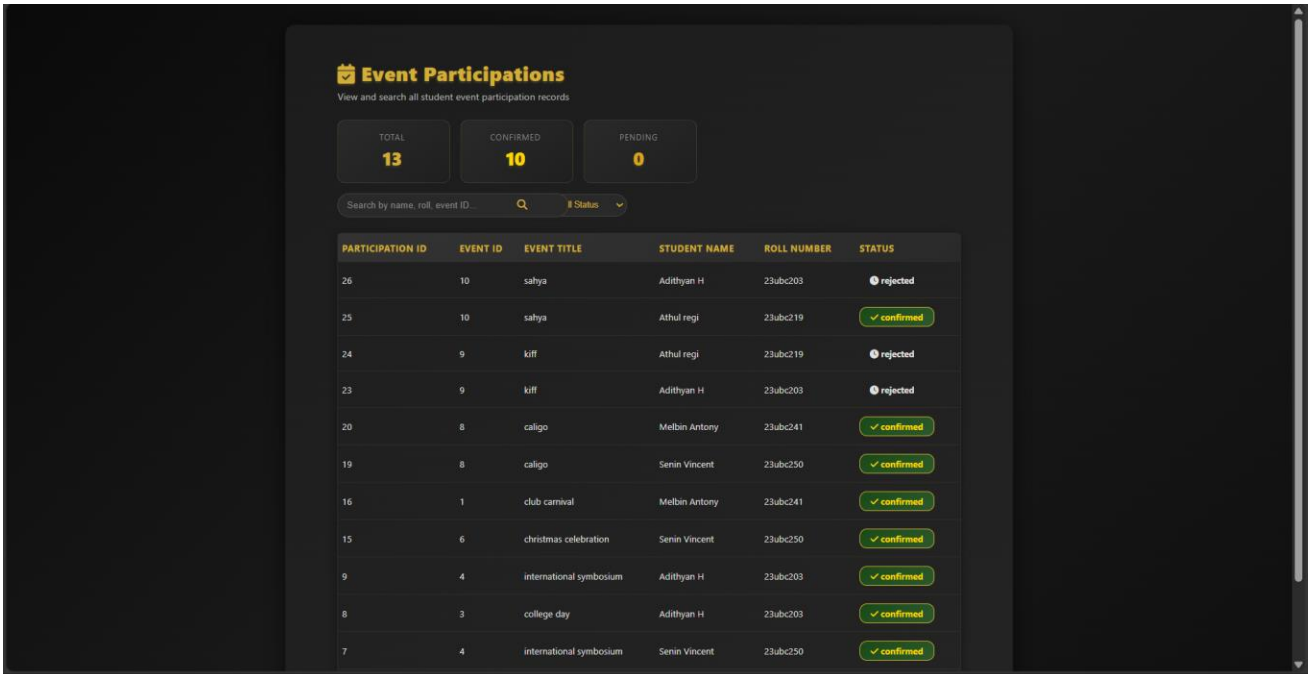


FACULTY MODULE

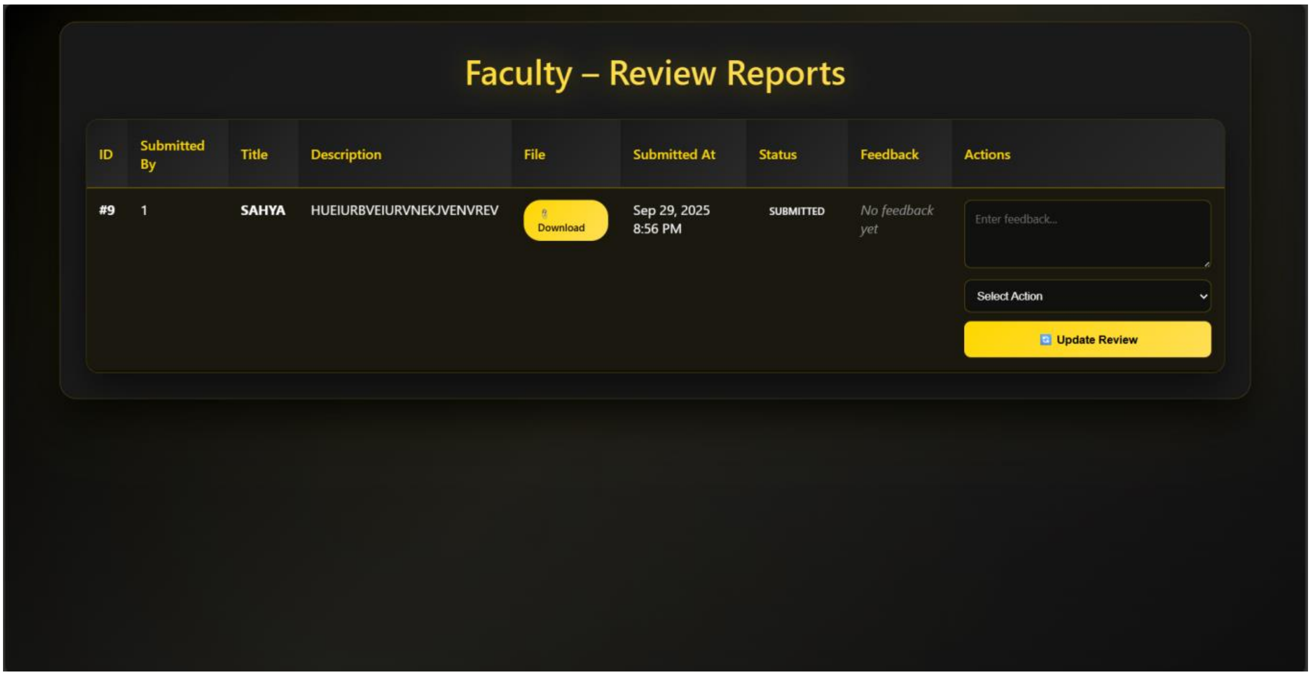
1. DASHBOARD PAGE



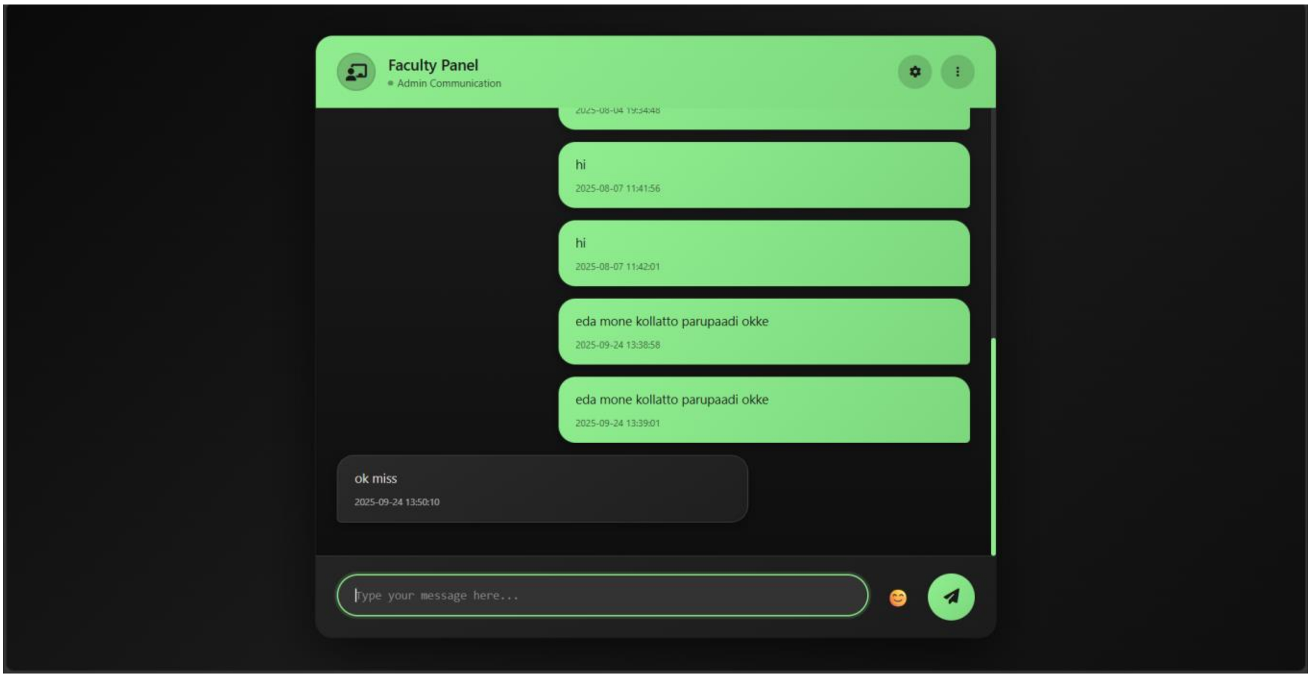
2. VIEW EVENT PARTICIPATION PAGE



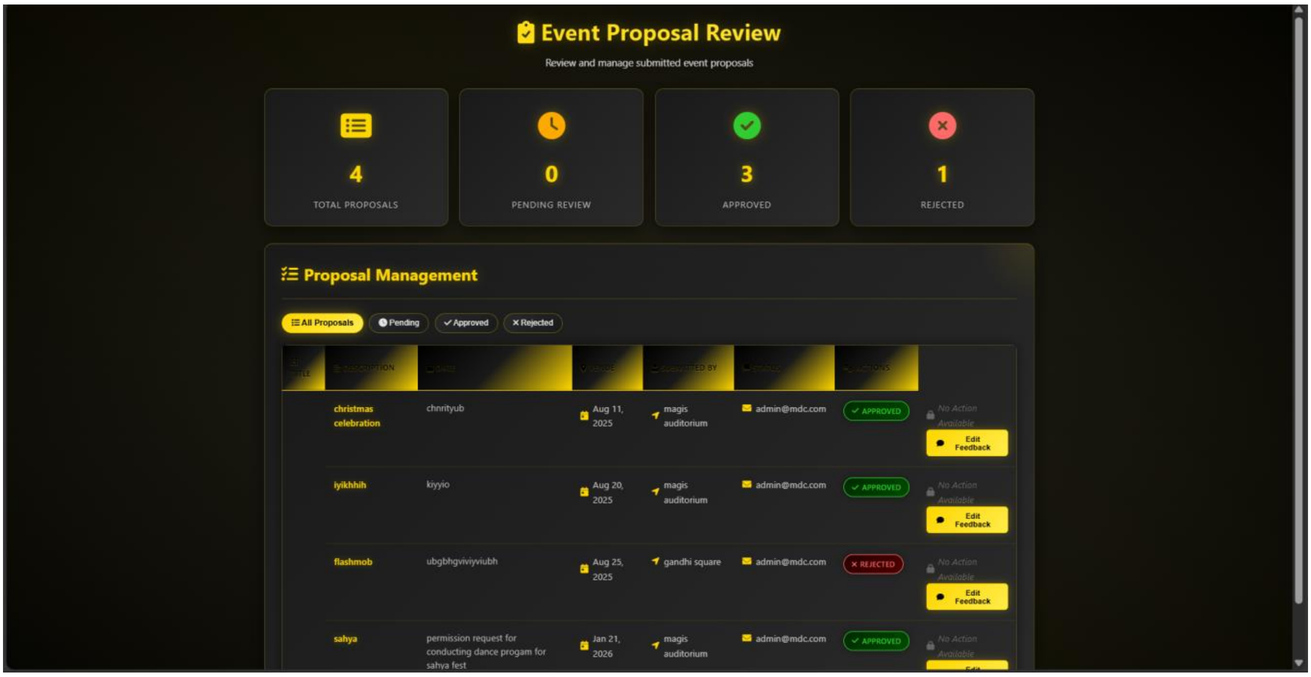
3. MANAGE REPORTS PAGE



4. CHAT WITH ADMIN PAGE



5. MANAGE EVENT PROPOSAL PAGE


Event Proposal Review
Review and manage submitted event proposals
4
TOTAL PROPOSALS
0
PENDING REVIEW
3
APPROVED
1
REJECTED
Proposal Management
All Proposals
Pending
Approved
Rejected
christmas celebration
chnrityub
Aug 11, 2025
magis auditorium
admin@mdc.com
APPROVED
No Action Available
Edit Feedback
iyikhhih
kiyyio
Aug 20, 2025
magis auditorium
admin@mdc.com
APPROVED
No Action Available
Edit Feedback
flashmob
ubgbhgviviyviubh
Aug 25, 2025
gandhi square
admin@mdc.com
REJECTED
No Action Available
Edit Feedback
sahya
permission request for conducting dance progam for sahya fest
Jan 21, 2026
magis auditorium
admin@mdc.com
APPROVED
No Action Available